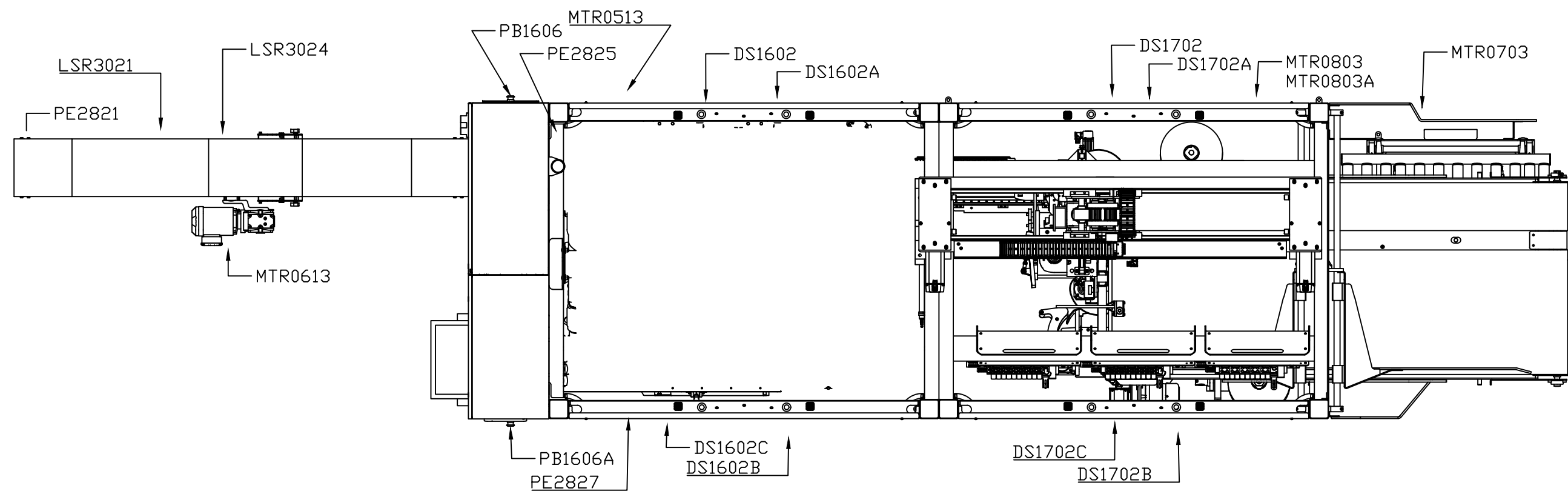


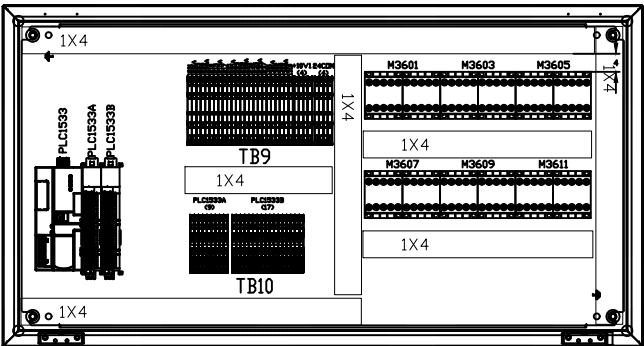
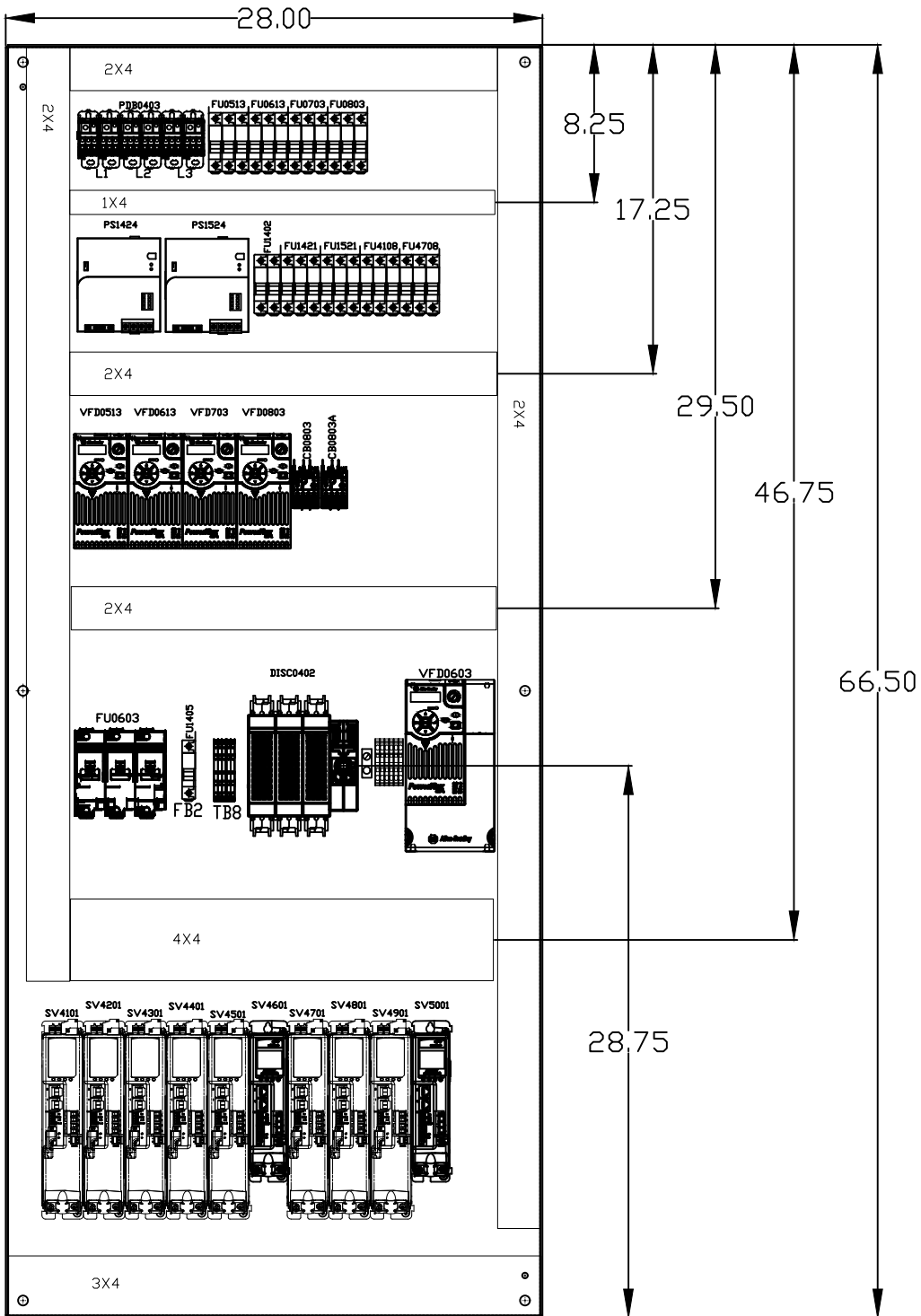
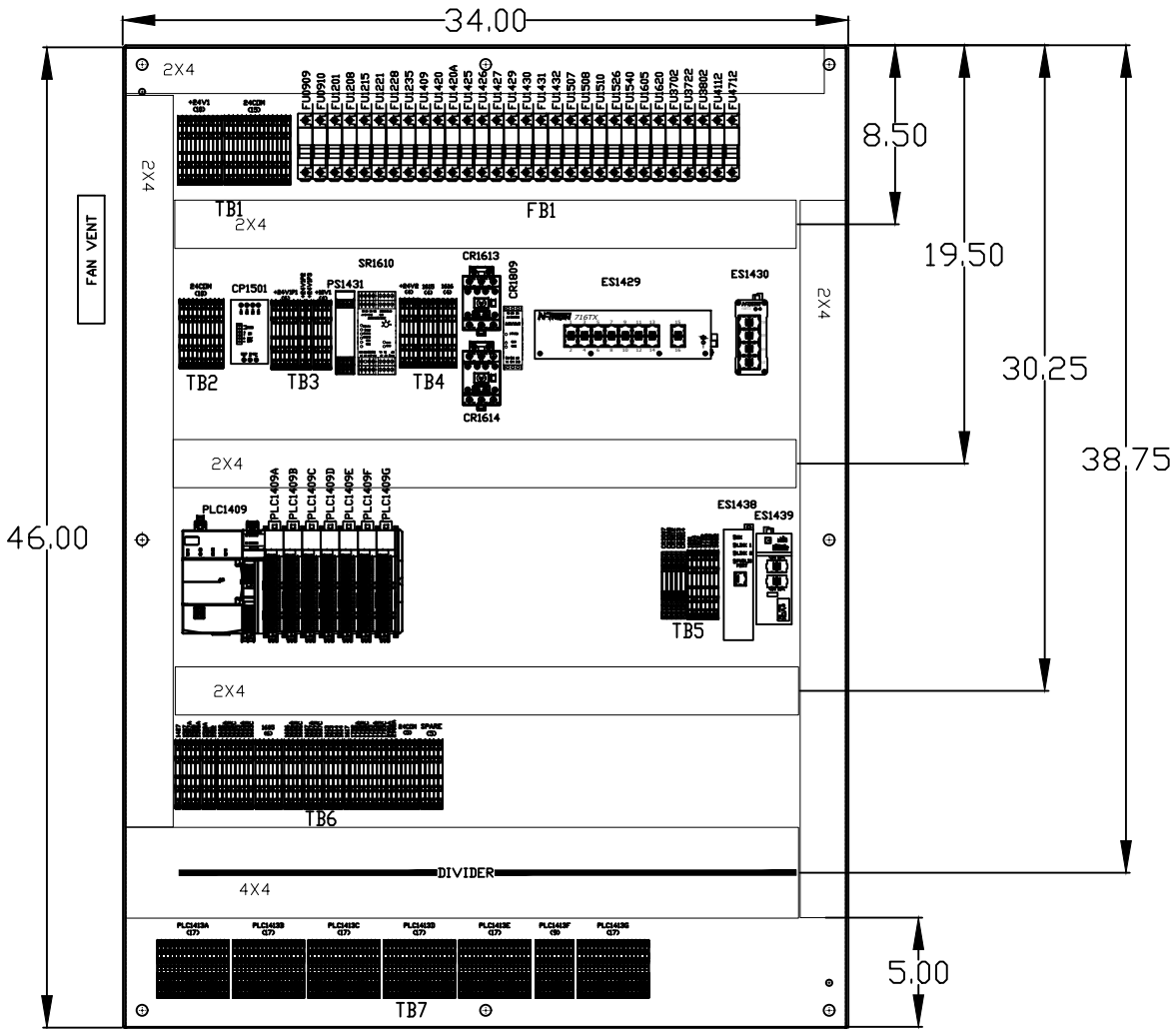
61546 Arte Sano DVPP w/ICE Electrical Schematics

BPA SCHEMATIC BUILD CONTENTS	
SHEET #	GENERIC SCHEMATIC LAYOUT DESCRIPTION
00	DRAWING INDEX AND PROJECT TITLE
01	SYSTEM LAYOUT
02	PANEL LAYOUT
03	TERMINAL LAYOUT
04	3 PHASE POWER DISTRIBUTION
05	VFD X 1
06	VFD X 2
07	VFD X 1
08	VFD X 1
09	DC INTERROLL MODULE
10 & 11	BLANK
12	LINEAR ACTUATORS X 6
13	LINEAR ACTUATORS X 2
14	3 PHASE TO 120VAC XFORMER, 120VAC & 24VDC DISTRIBUTION
15	24VDC CONT.
16	SAFETY CIRCUIT - [CAT 3]
17	SAFETY CIRCUIT - [CAT 3] CONT.
18	AIR DUMP CIRCUIT
19 - 27	BLANK
28	PLC DIGITAL INPUTS X 2
29	PLC DIGITAL INPUTS X 2
30	PLC DIGITAL INPUTS X 1. ANALOG INPUTS X 1
31	PLC ANALOG INPUTS X 1
32	BLANK
35	PLC OUTPUTS X 1
36	PLC OUTPUTS X 1
37	ETHERNET VALVE BANK X 2
38	ETHERNET VALVE BANK X 1
39 - 41	BLANK
41	SERVO #1 ROBOT AXIS 1
42	SERVO #2 ROBOT AXIS 2
43	SERVO #3 ROBOT AXIS 3
44	SERVO #4 LOADING RAM
45	SERVO #5 INDEXING BELT
46	SERVO #6 CASSETTE ACTIVE WALL
47	SERVO #7 ICE X-AXIS
48	SERVO #8 ICE Y-AXIS
49	SERVO #9 TIPPER
50	SERVO #10 CASE OPENING ARM
51	ETHERNET NETWORK
52 - 54	BLANK
55	TORQUE SPECIFICATIONS

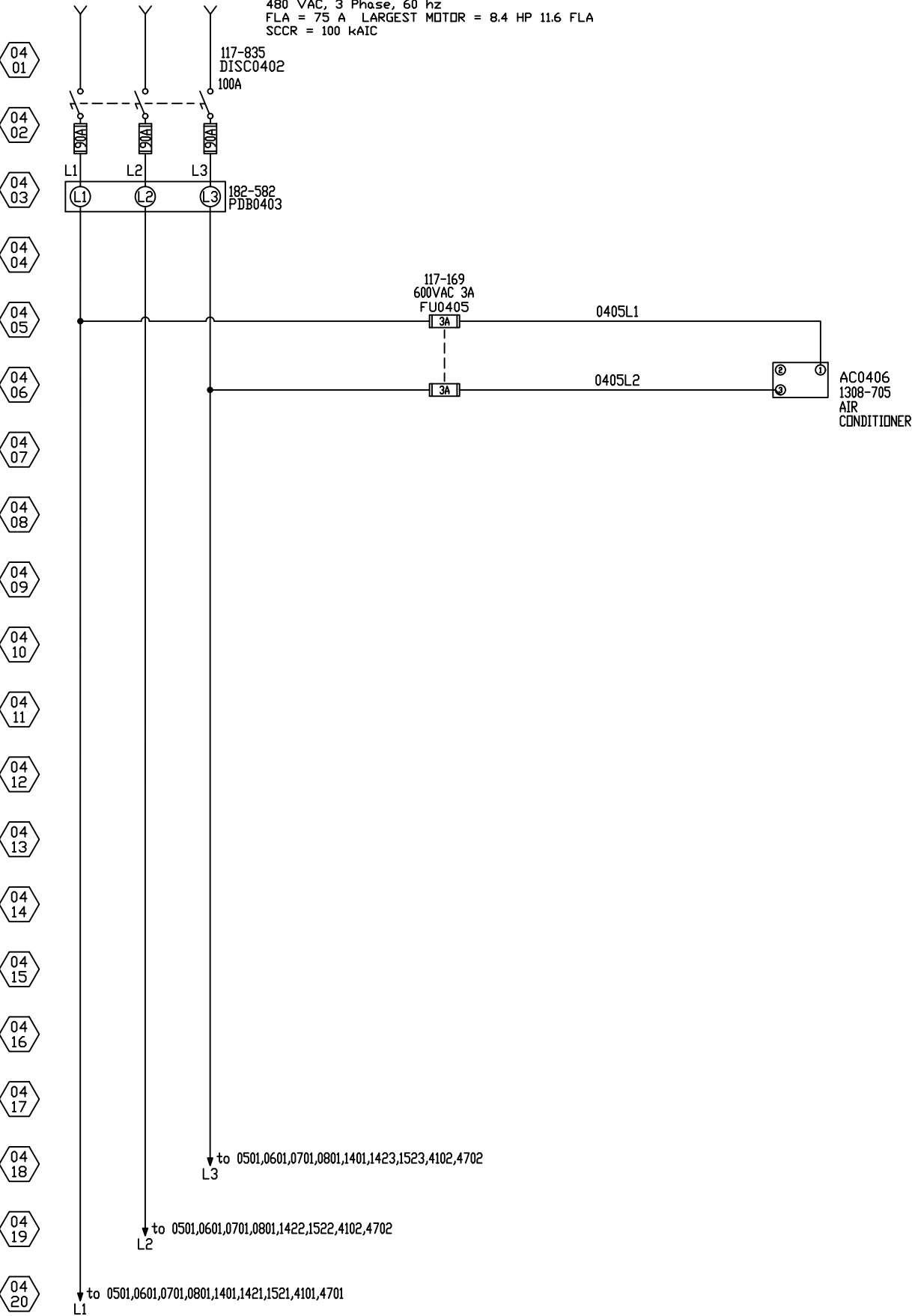
61546 Arte Sano DVPP w/ICE Device Layout



61546 Arte Sano Panel Layout

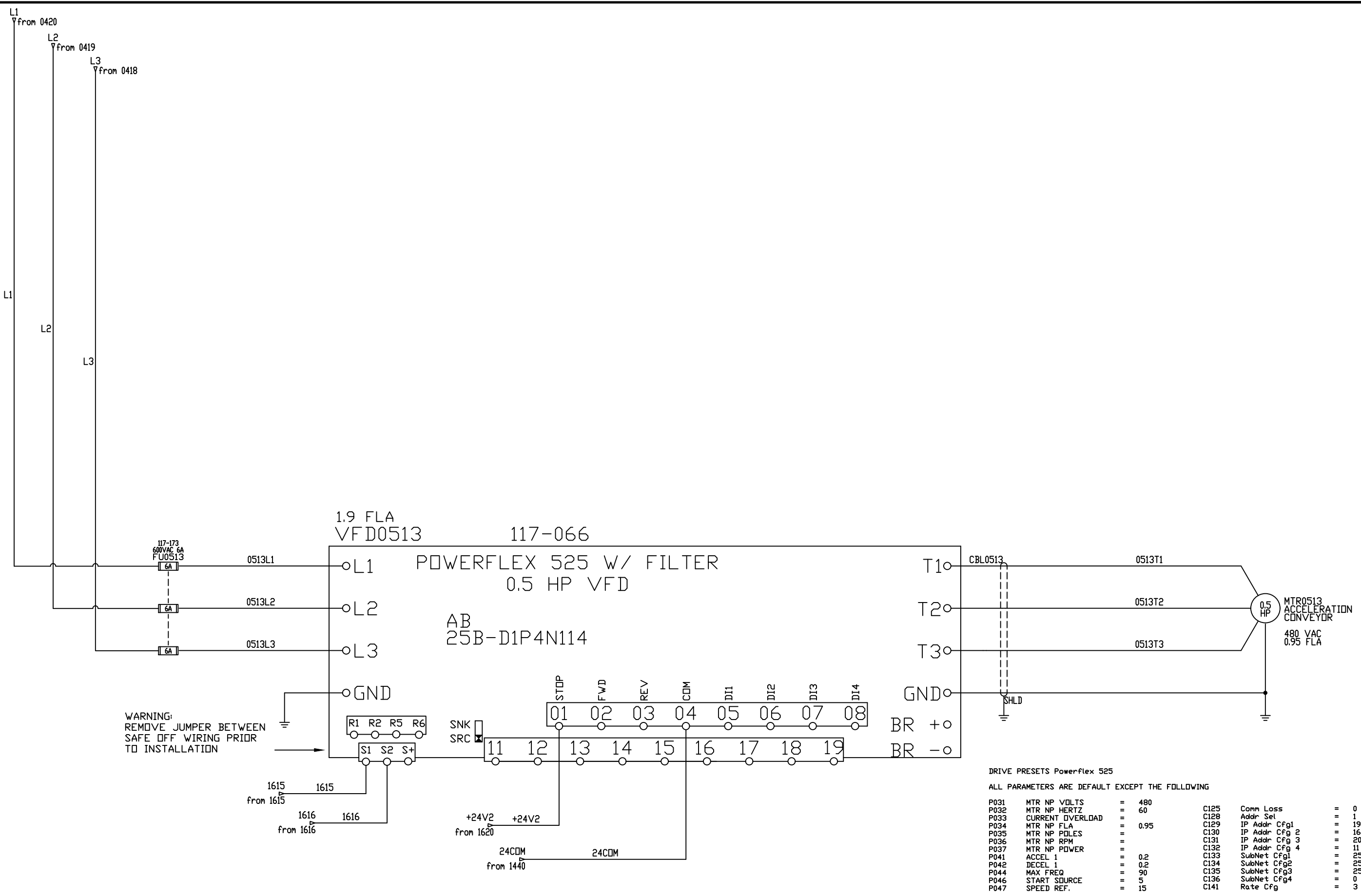






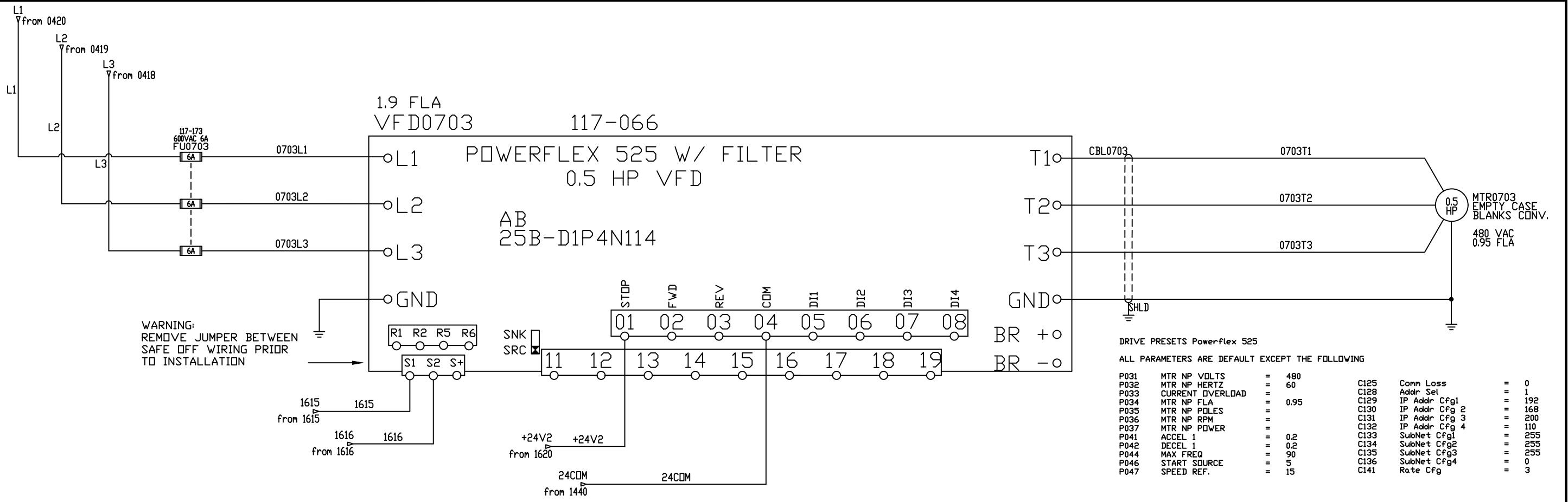
- 04 21
- 04 22
- 04 23
- 04 24
- 04 25
- 04 26
- 04 27
- 04 28
- 04 29
- 04 30
- 04 31
- 04 32
- 04 33
- 04 34
- 04 35
- 04 36
- 04 37
- 04 38
- 04 39
- 04 40

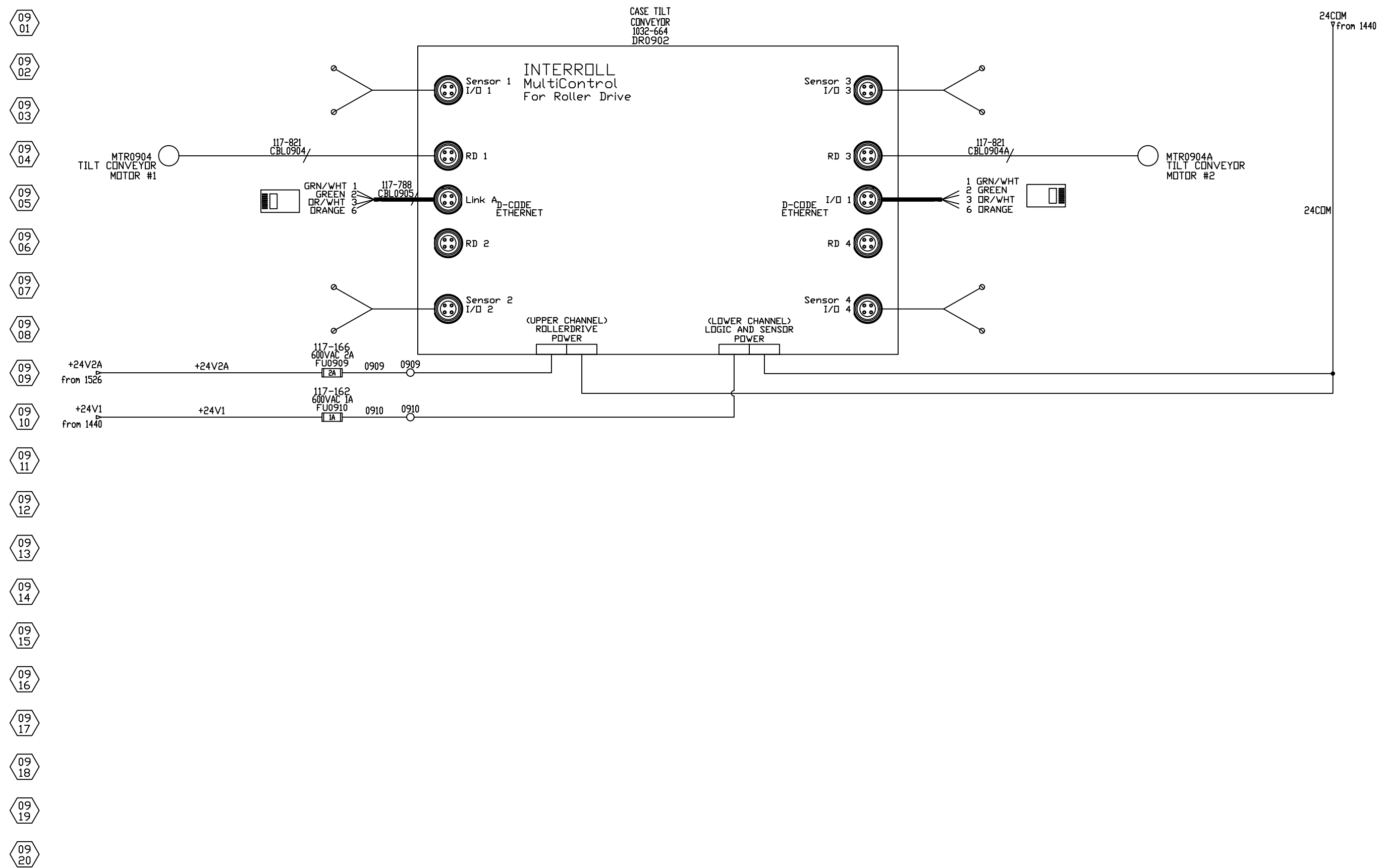
- 05 01
- 05 02
- 05 03
- 05 04
- 05 05
- 05 06
- 05 07
- 05 08
- 05 09
- 05 10
- 05 11
- 05 12
- 05 13
- 05 14
- 05 15
- 05 16
- 05 17
- 05 18
- 05 19
- 05 20



ALL RIGHTS RESERVED, DRAWING REMAINS PROPERTY OF BLUEPRINT AUTOMATION

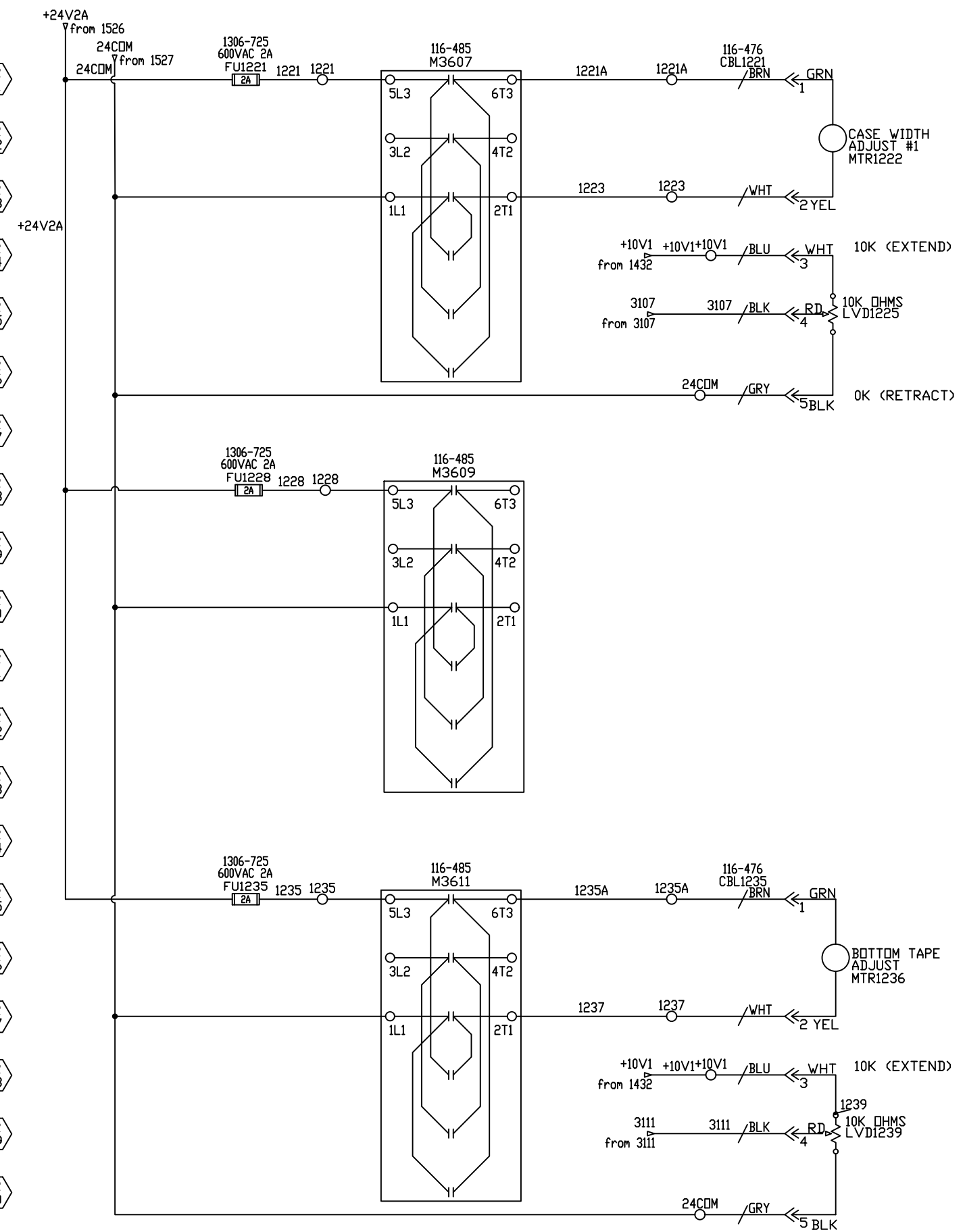
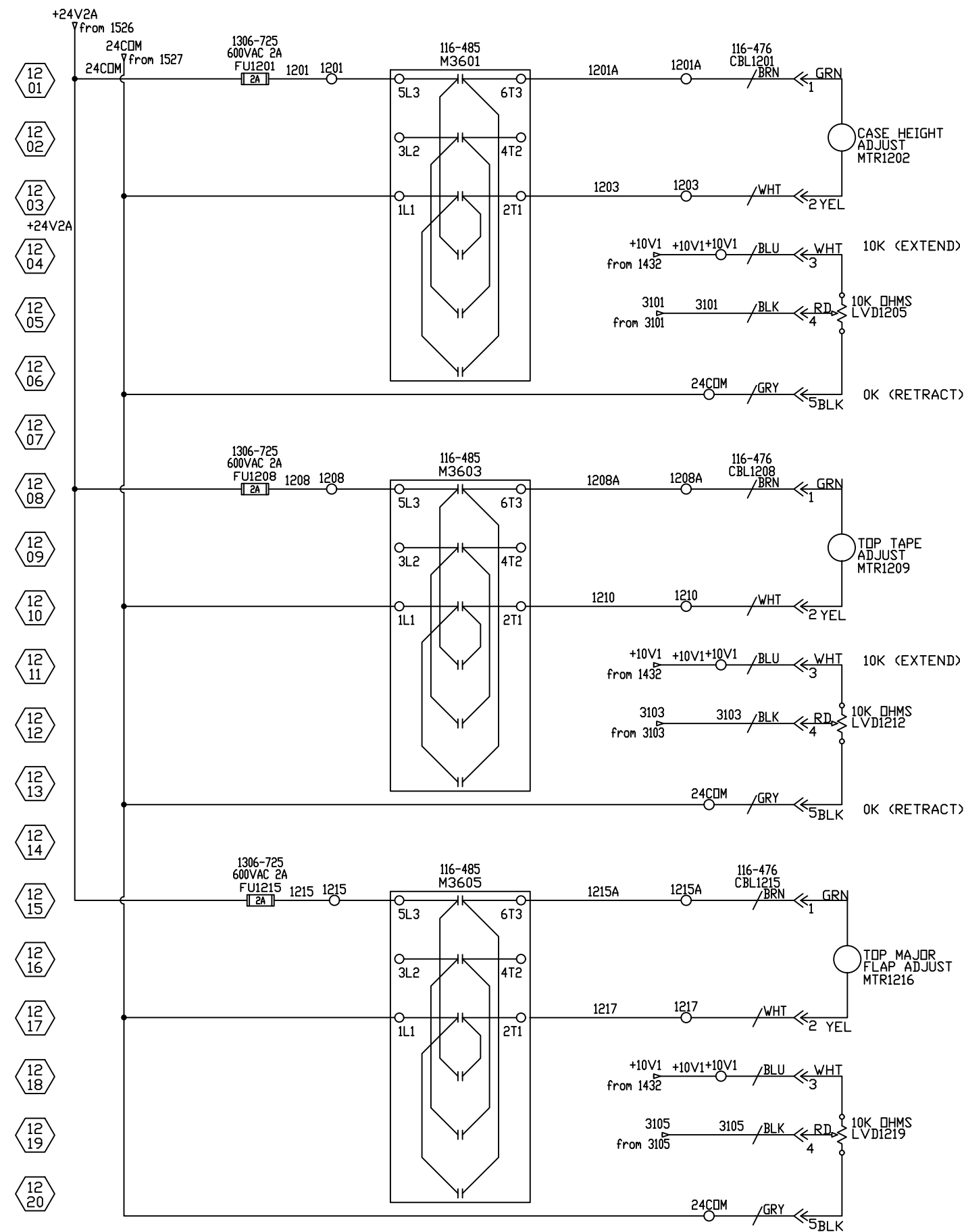
07 01
07 02
07 03
07 04
07 05
07 06
07 07
07 08
07 09
07 10
07 11
07 12
07 13
07 14
07 15
07 16
07 17
07 18
07 19
07 20





- 1001
- 1002
- 1003
- 1004
- 1005
- 1006
- 1007
- 1008
- 1009
- 1010
- 1011
- 1012
- 1013
- 1014
- 1015
- 1016
- 1017
- 1018
- 1019
- 1020

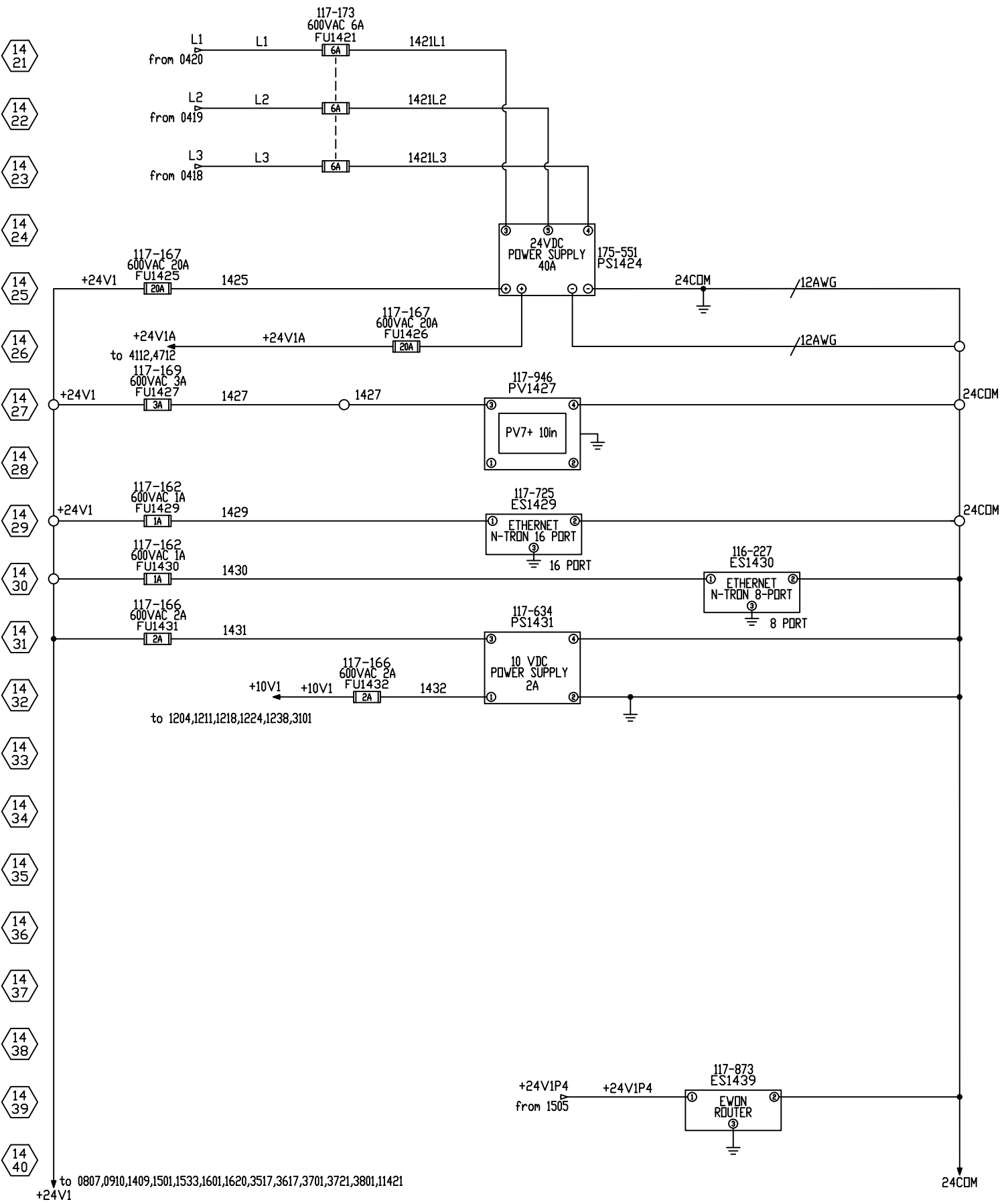
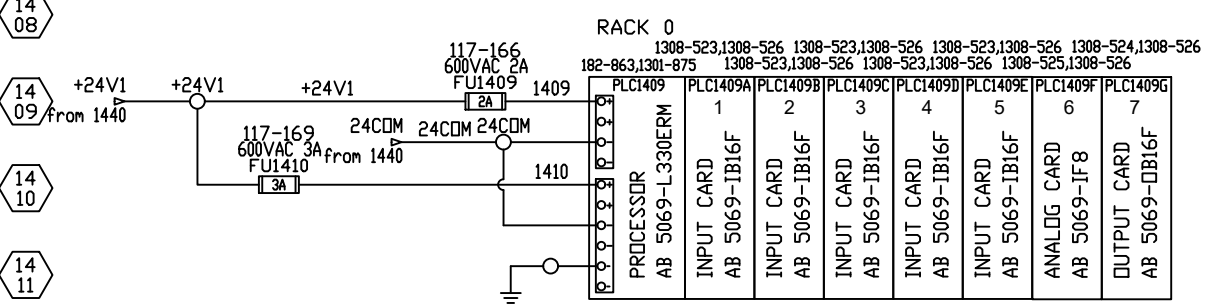
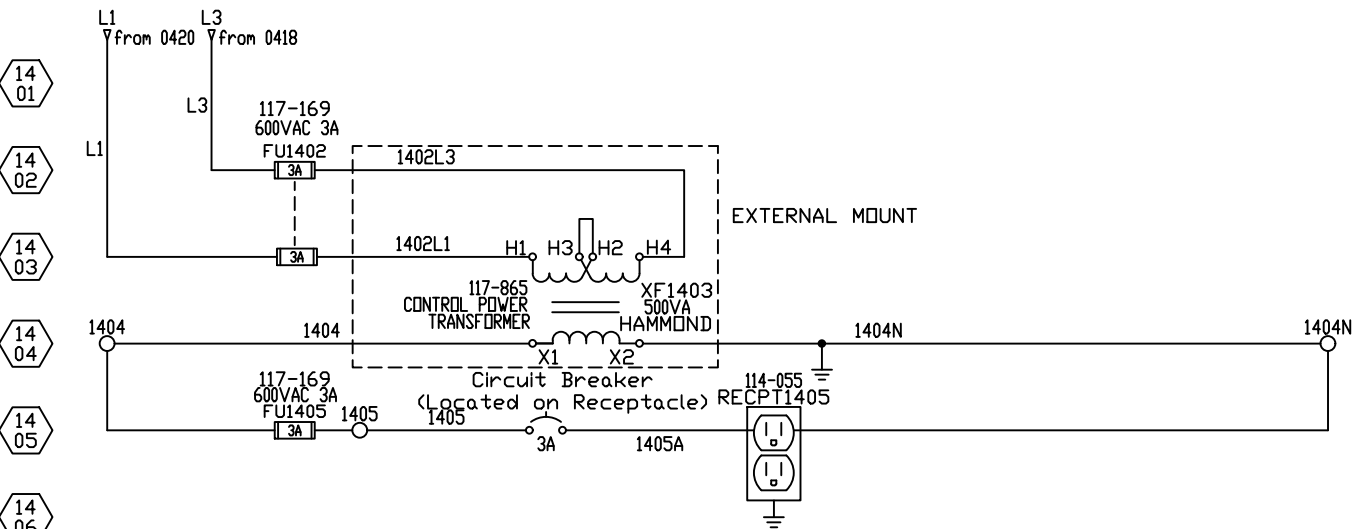
PAGES 10 & 11 WERE INTENTIONALLY LEFT BLANK

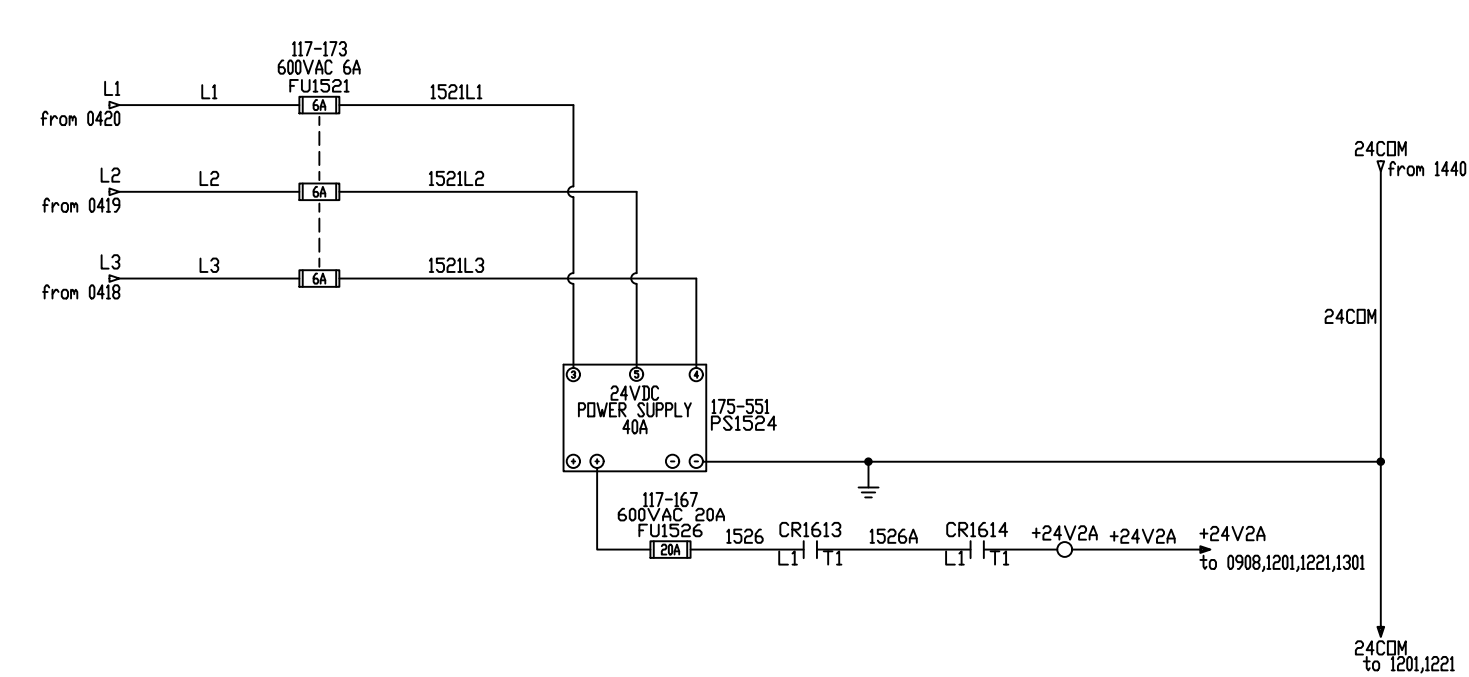
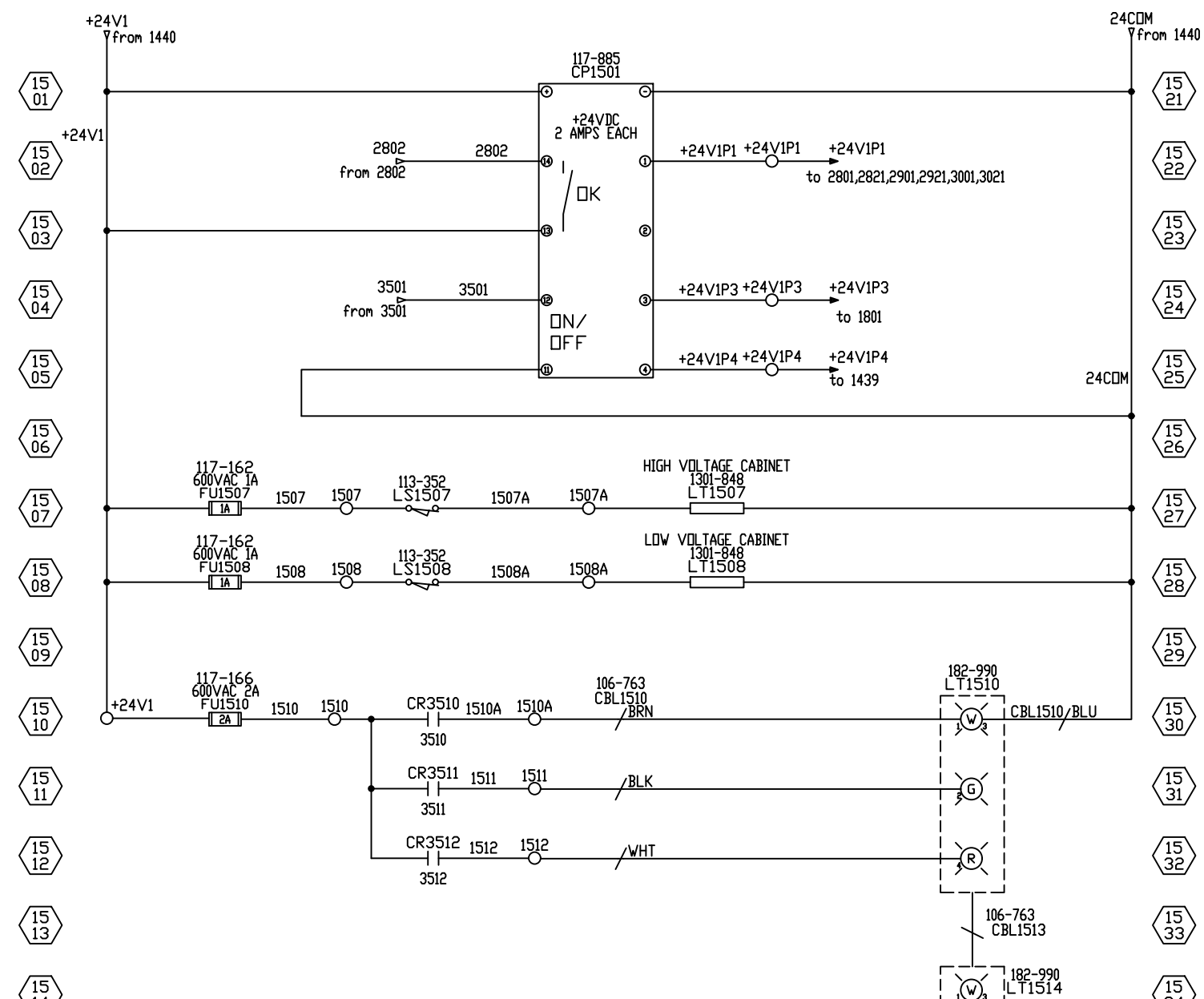


13 01
13 02
13 03
13 04
13 05
13 06
13 07
13 08
13 09
13 10
13 11
13 12
13 13
13 14
13 15
13 16
13 17
13 18
13 19
13 20

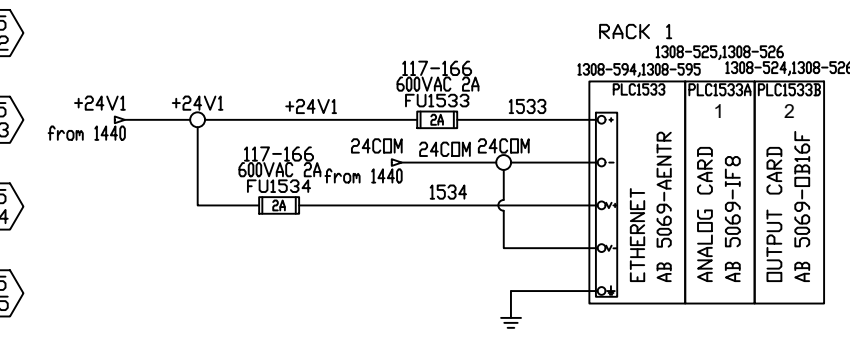
PAGE 13 WAS INTENTIONALLY LEFT BLANK

13 21
13 22
13 23
13 24
13 25
13 26
13 27
13 28
13 29
13 30
13 31
13 32
13 33
13 34
13 35
13 36
13 37
13 38
13 39
13 40

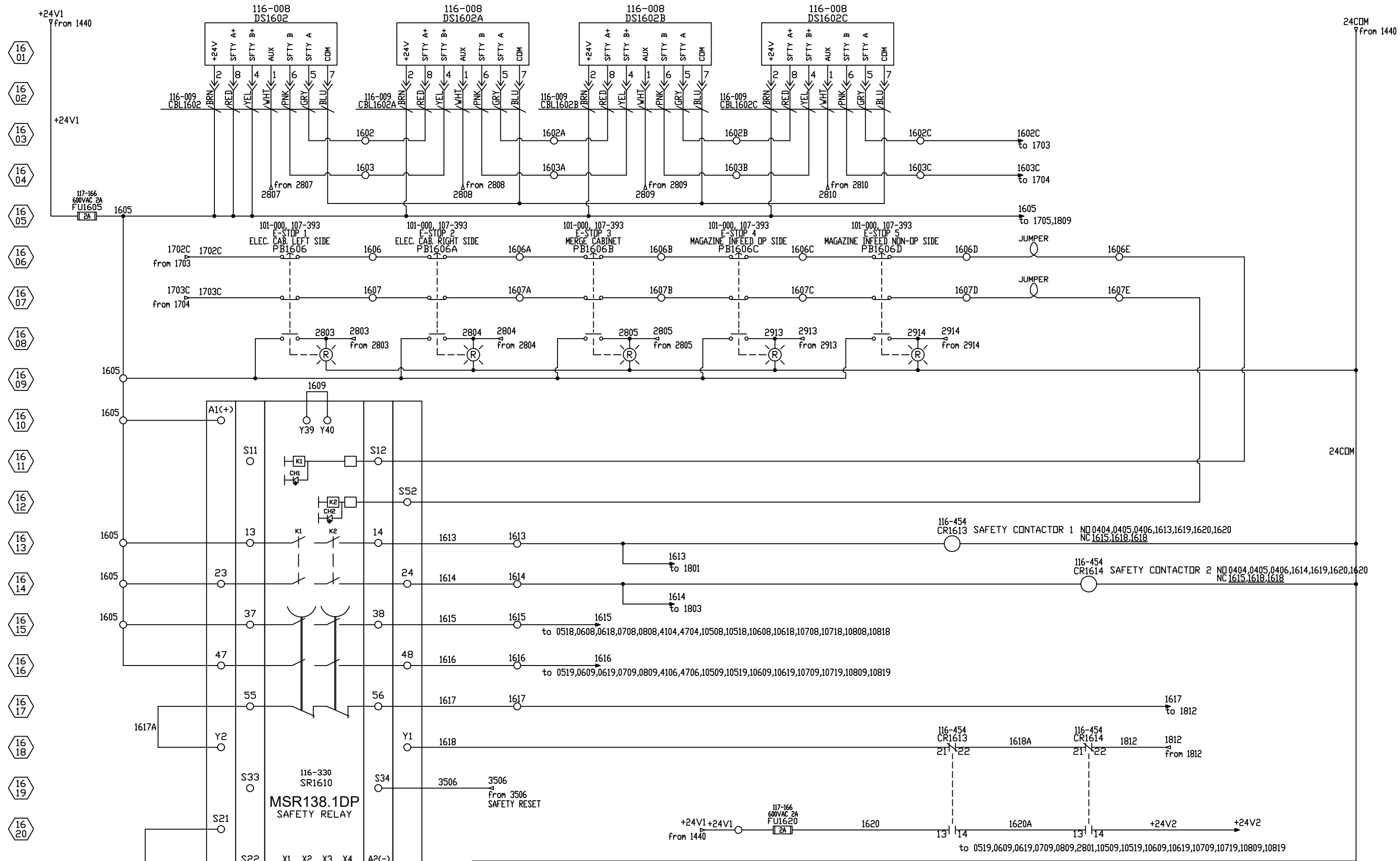




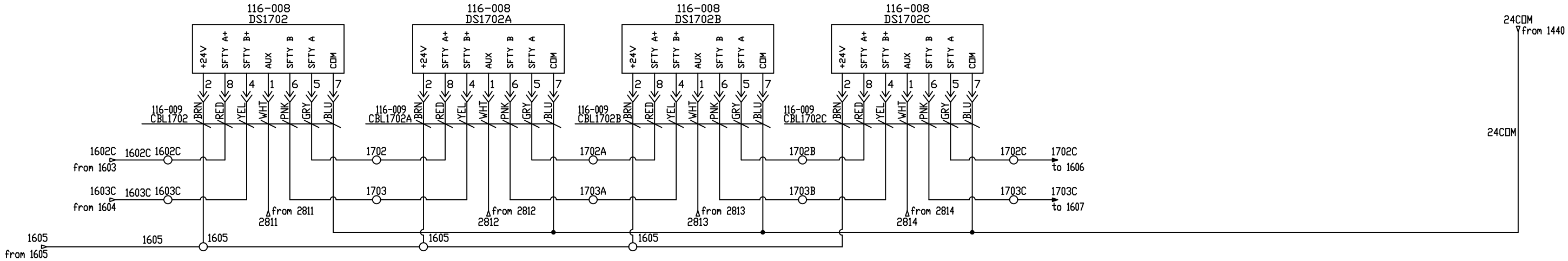
3 Color Override Control (Color 3 overrides Colors 1 and 2, Color 2 overrides Color 1)			
Input 1: Pin 1 Brown Wire	Input 2: Pin 4 Black Wire	Input 3: Pin 2 White Wire	LED Color
—	—	—	Light Off
+24 V dc	—	—	Color 1 On
—	+24 V dc	—	Color 2 On
+24 V dc	+24 V dc	—	Color 2 On
—	—	+24 V dc	Color 3 On
+24 V dc	—	+24 V dc	Color 3 On
—	+24 V dc	+24 V dc	Color 3 On
+24 V dc	+24 V dc	+24 V dc	Color 3 On

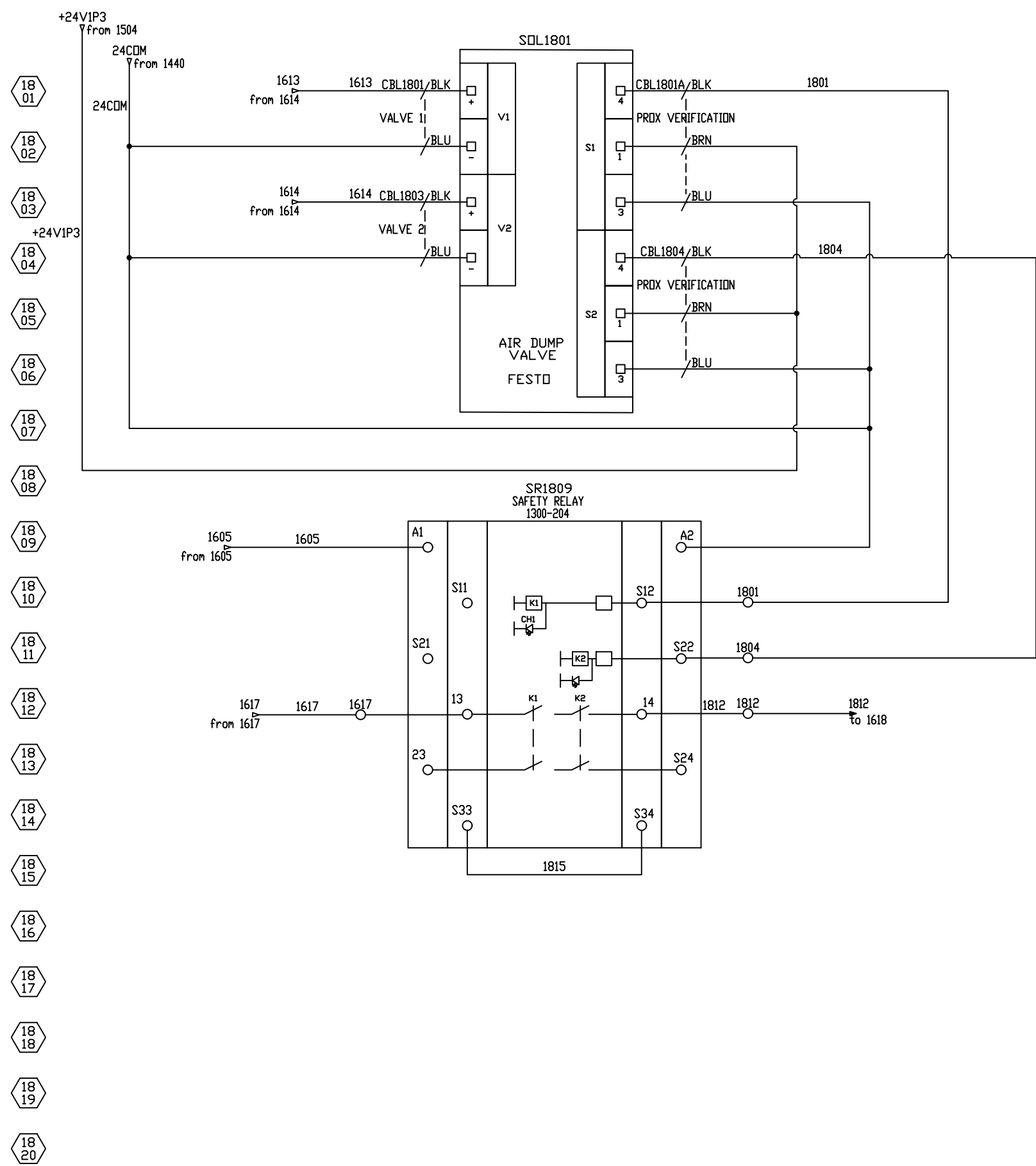


ALL RIGHTS RESERVED, DRAWING REMAINS PROPERTY OF BLUEPRINT AUTOMATION



17 01
17 02
17 03
17 04
17 05
17 06
17 07
17 08
17 09
17 10
17 11
17 12
17 13
17 14
17 15
17 16
17 17
17 18
17 19
17 20

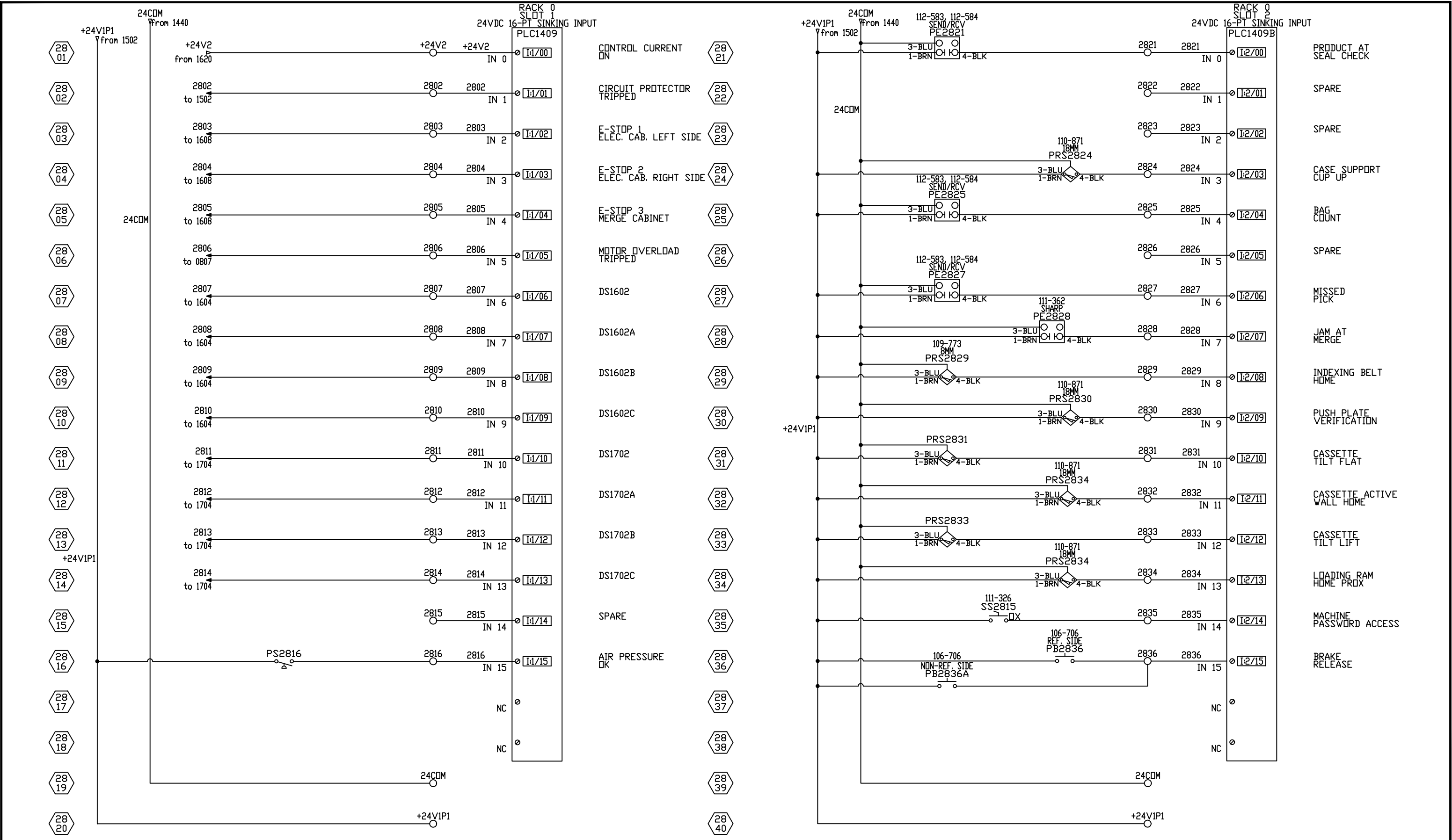


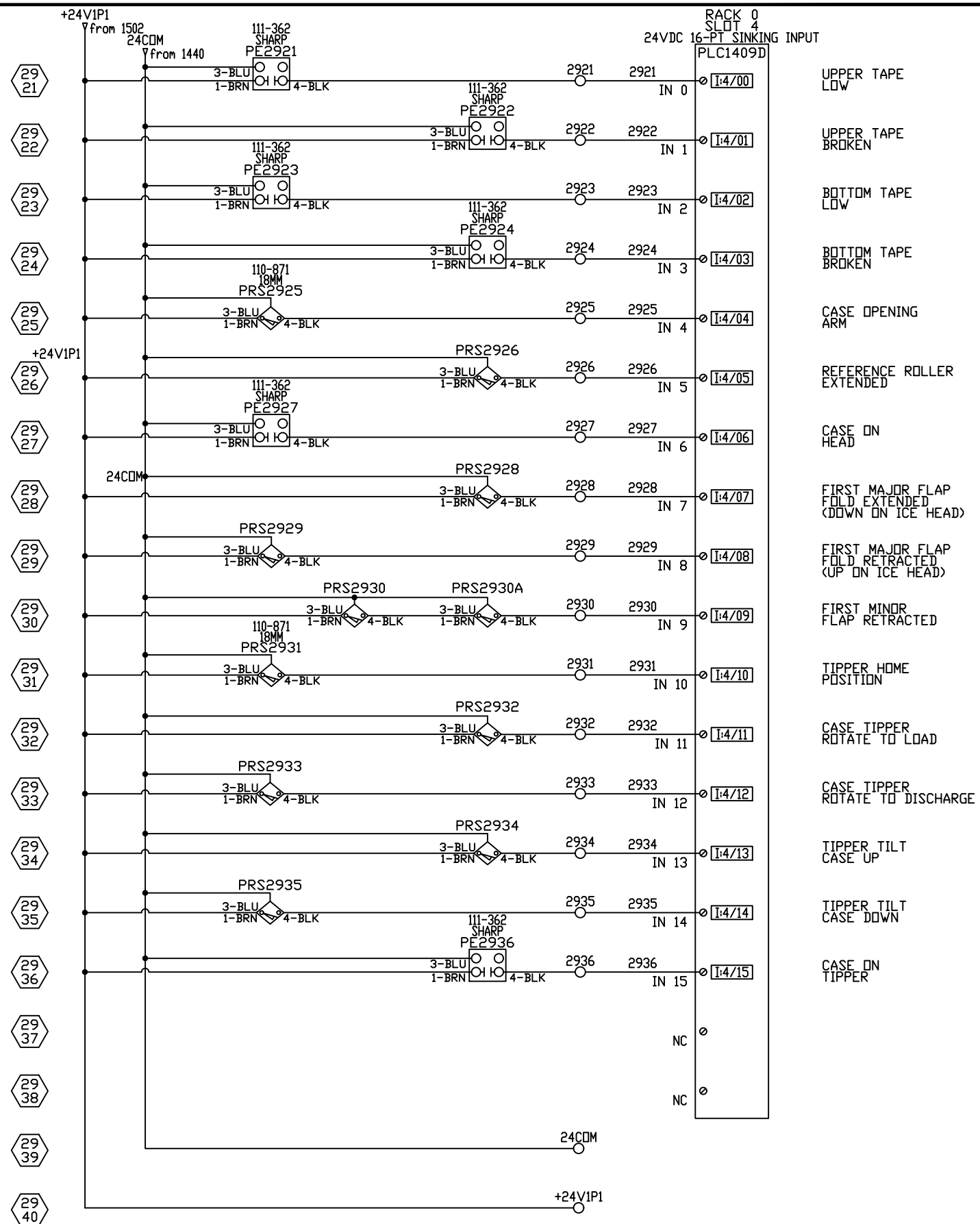


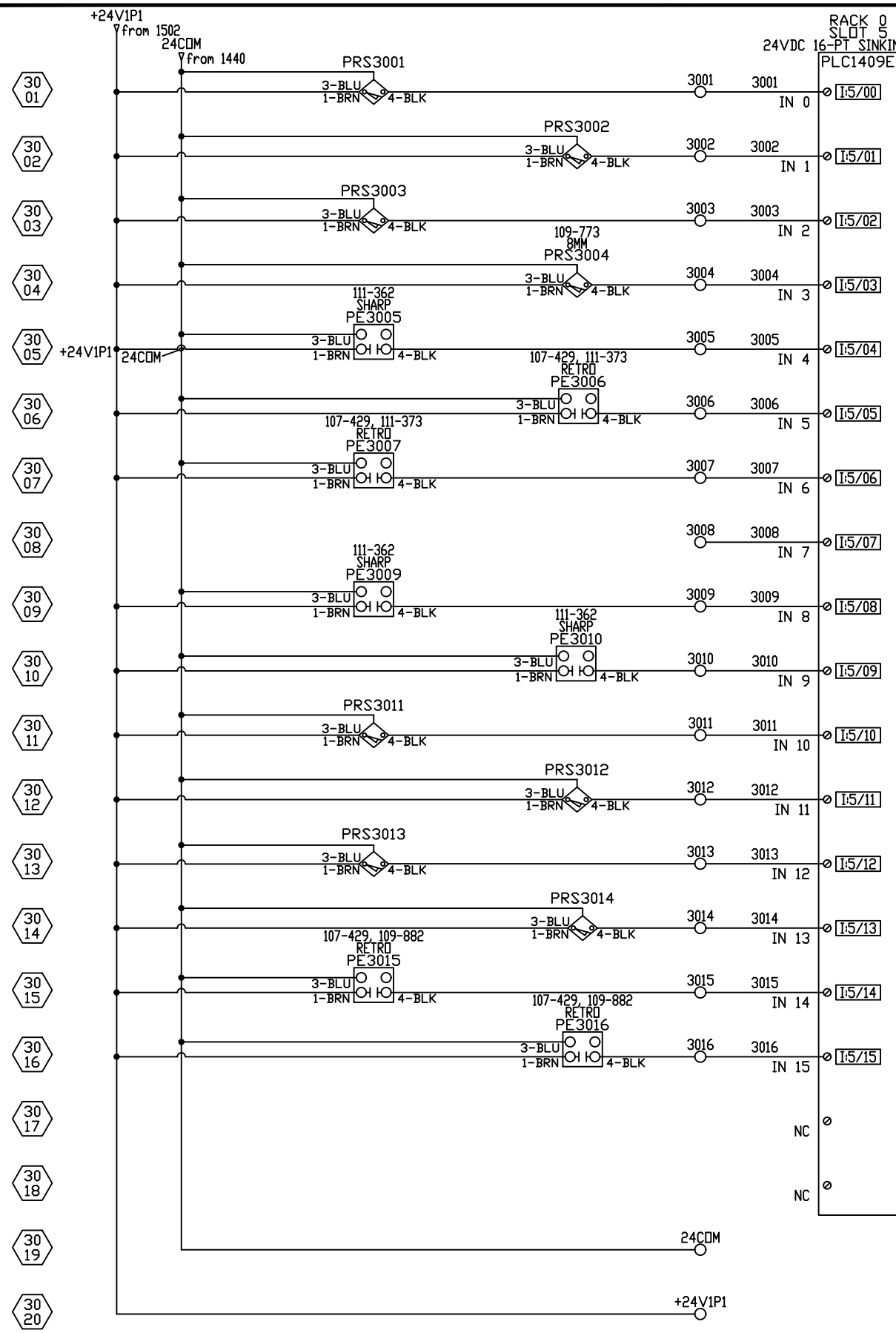
18 01
18 02
18 03
18 04
18 05
18 06
18 07
18 08
18 09
18 10
18 11
18 12
18 13
18 14
18 15
18 16
18 17
18 18
18 19
18 20

- 1901
- 1902
- 1903
- 1904
- 1905
- 1906
- 1907
- 1908
- 1909
- 1910
- 1911
- 1912
- 1913
- 1914
- 1915
- 1916
- 1917
- 1918
- 1919
- 1920

PAGES 19 - 27 WERE INTENTIONALLY LEFT BLANK







TIPPER FLAP
TUCKER CLOSE
(CLOSED)

TIPPER FLAP
TUCKER OPEN

(OPEN)
MINOR KICKER
RETRACTED

ROTATION ARM
POSITION FLAG

CASE CLEAR TO
TUCK MAJOR FLAP

CASE AT TOP
MAJOR FLAP TUCK

CASE AT BOTTOM
MAJOR FLAP TUCK

SPARE

CASE PRESENT
TO TILT

CASE CLEAR
TO TILT

TILT CONVEYOR
IN RECEIVE POS.

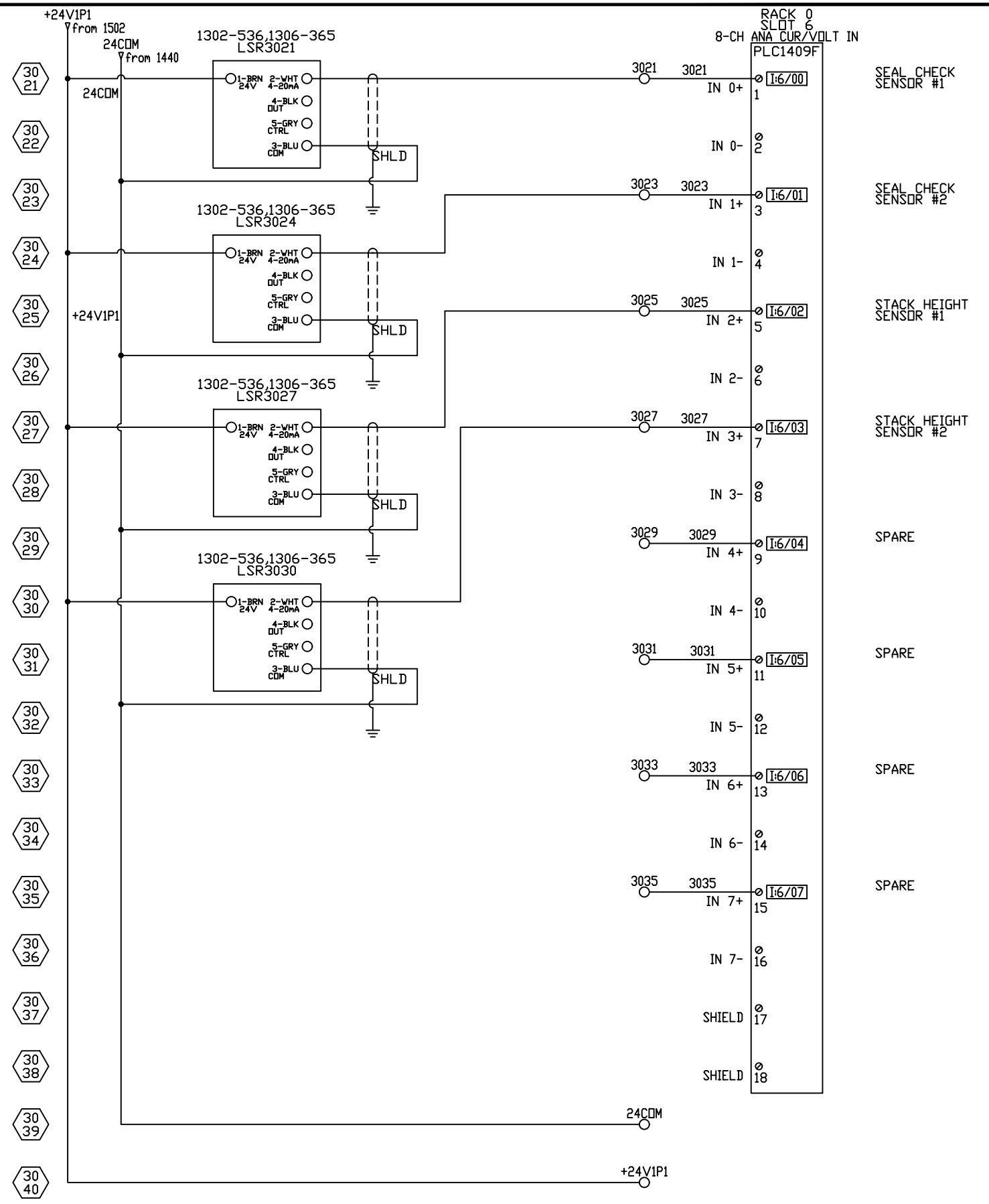
TILT CONVEYOR
RELEASE POS.

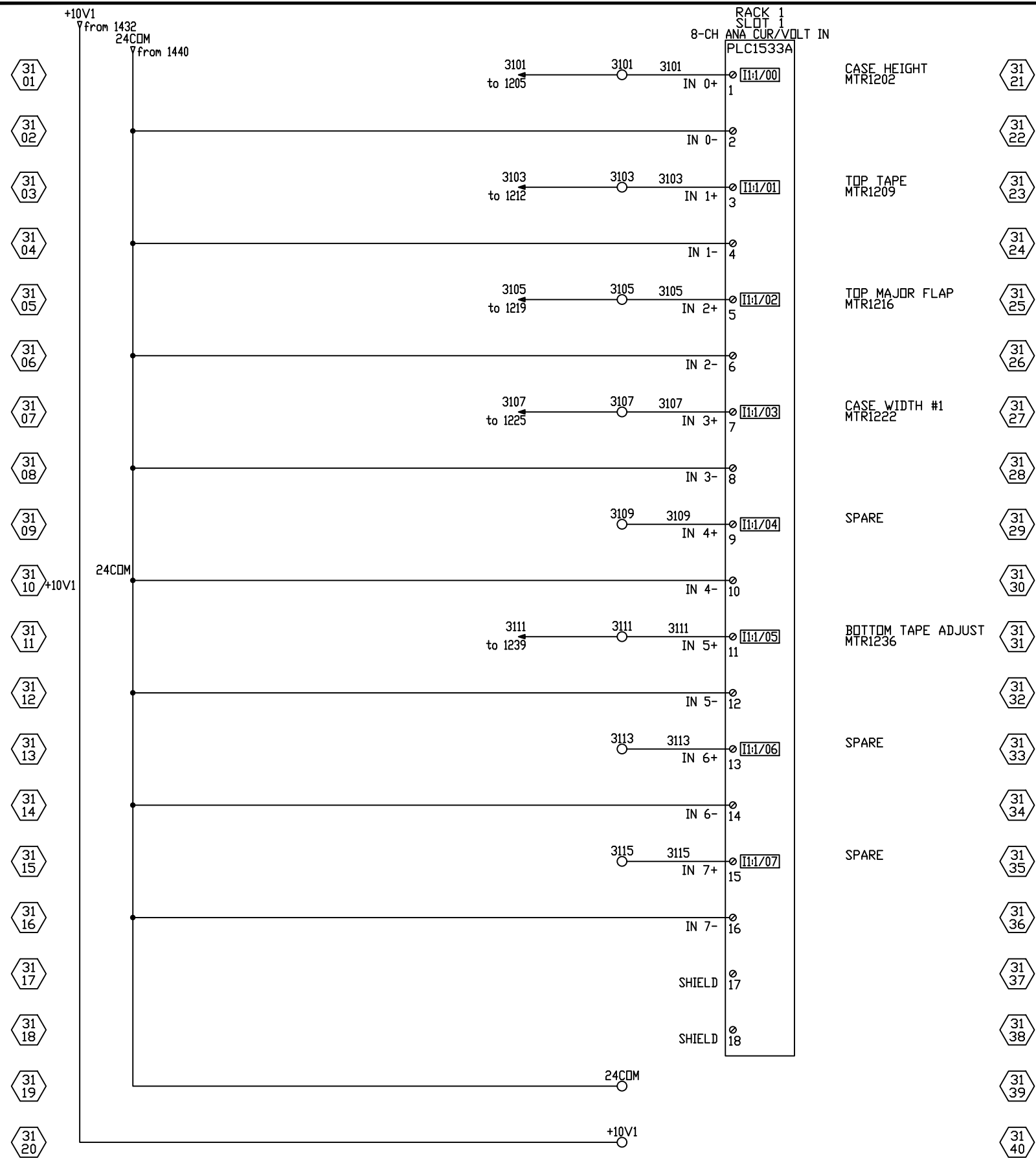
TILT FINGERS IN
CASE RECEIVE POS.

TILT FINGERS IN
CASE RELEASE POS.

FULL CASE
EXIT CLEAR

FULL CASE
EXIT BACK-UP





32
01

32
02

32
03

32
04

32
05

32
06

32
07

32
08

32
09

32
10

32
11

32
12

32
13

32
14

32
15

32
16

32
17

32
18

32
19

32
20

PAGES 32 - 34 WERE INTENTIONALLY LEFT BLANK

32
21

32
22

32
23

32
24

32
25

32
26

32
27

32
28

32
29

32
30

32
31

32
32

32
33

32
34

32
35

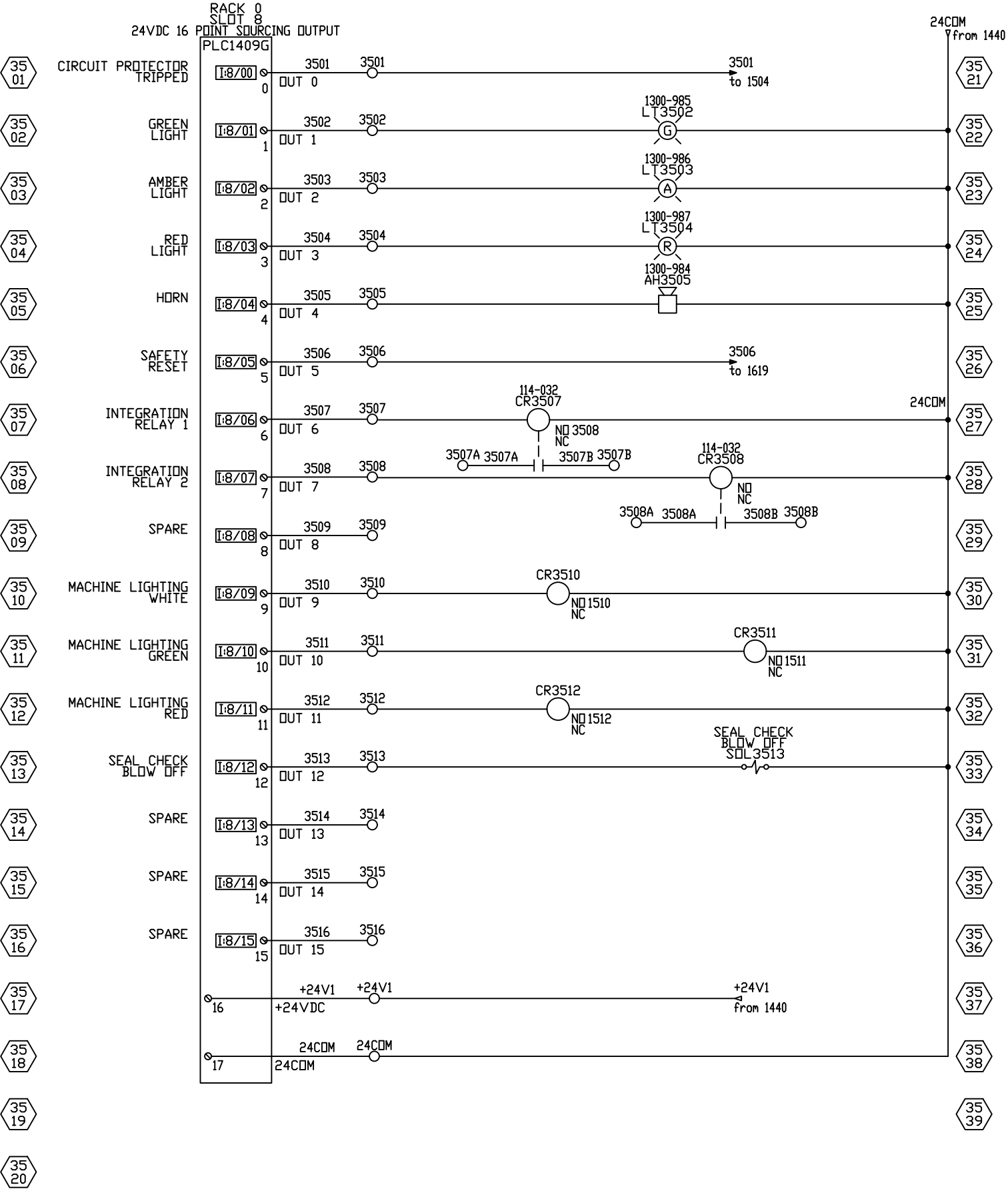
32
36

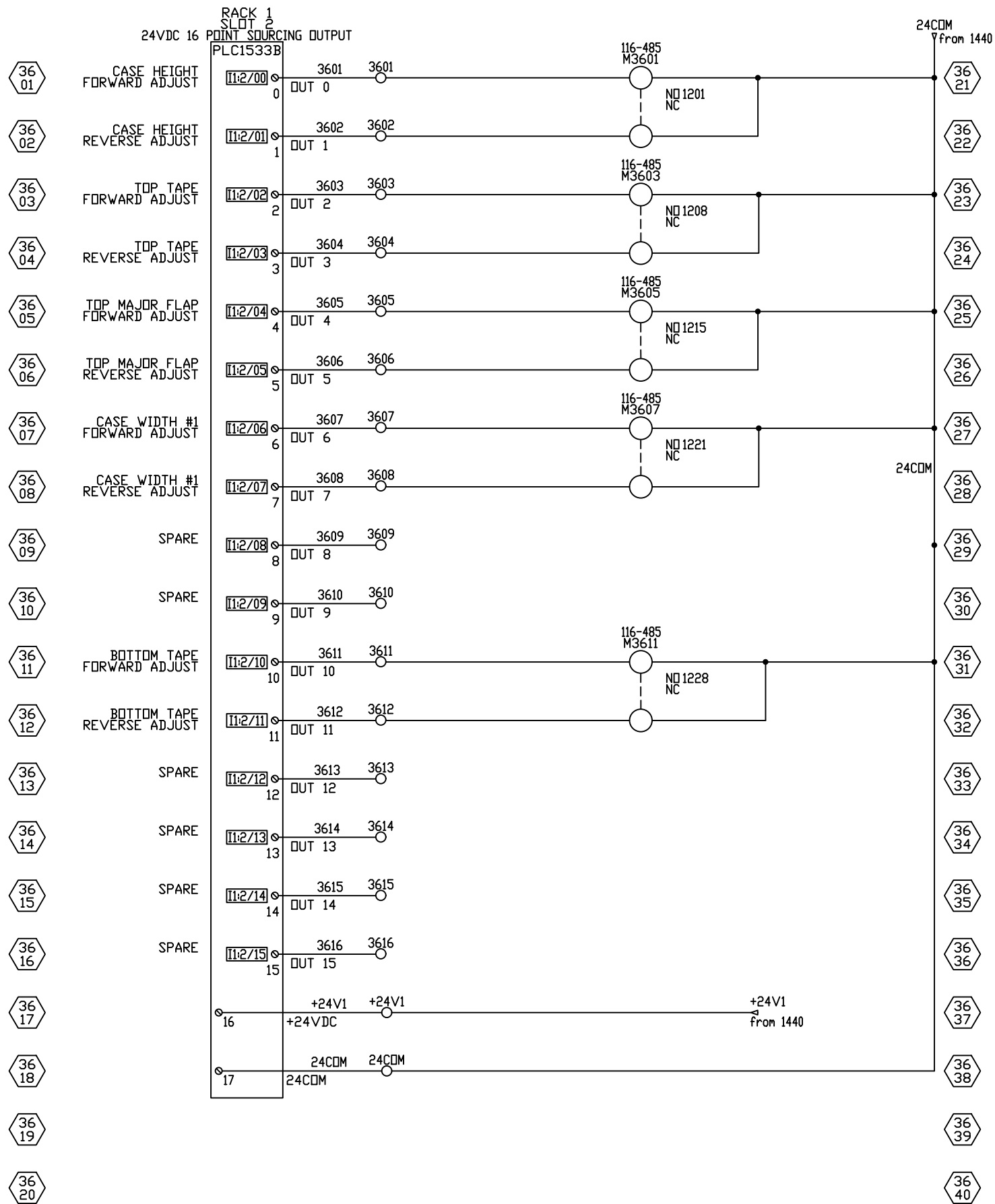
32
37

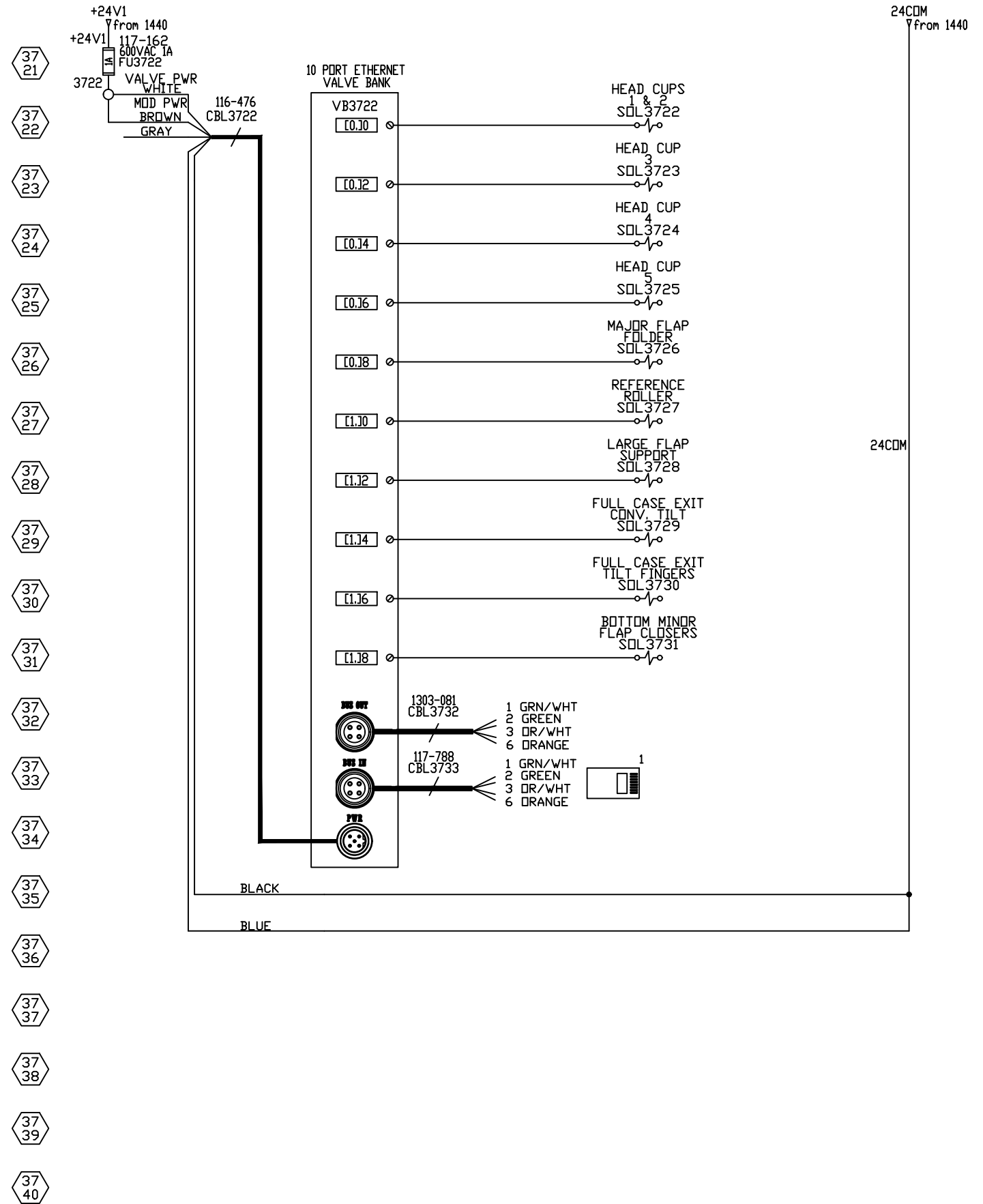
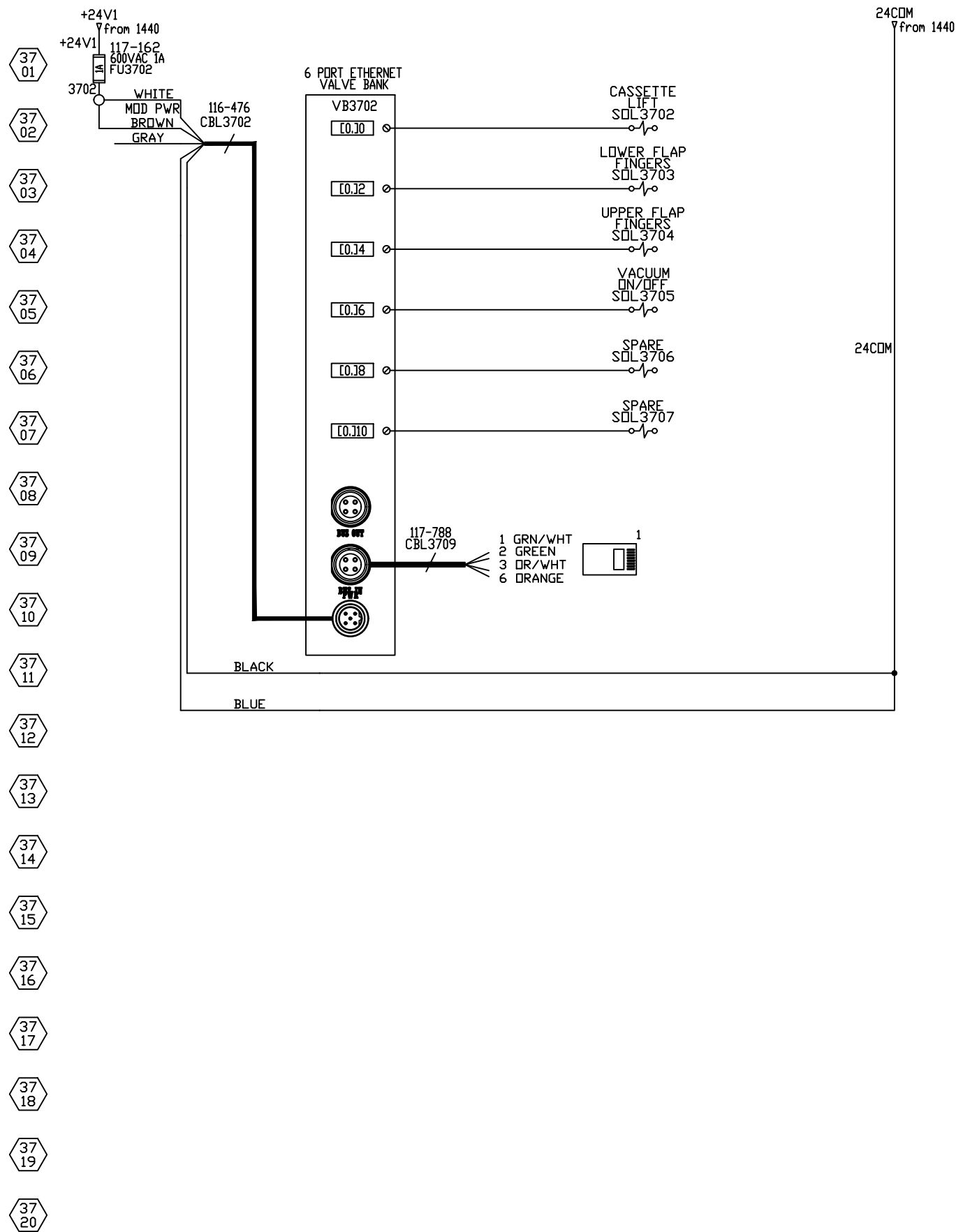
32
38

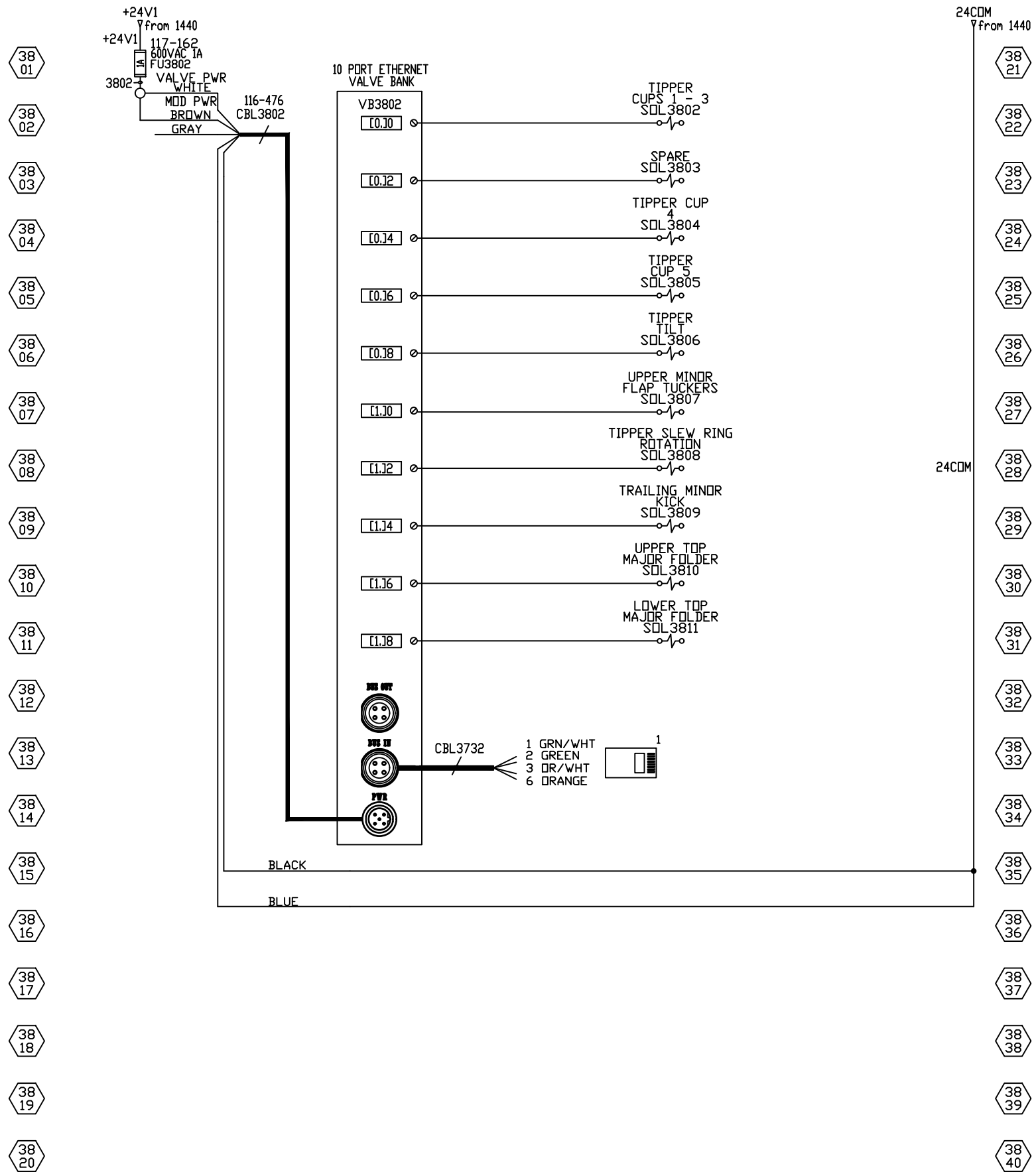
32
39

32
40







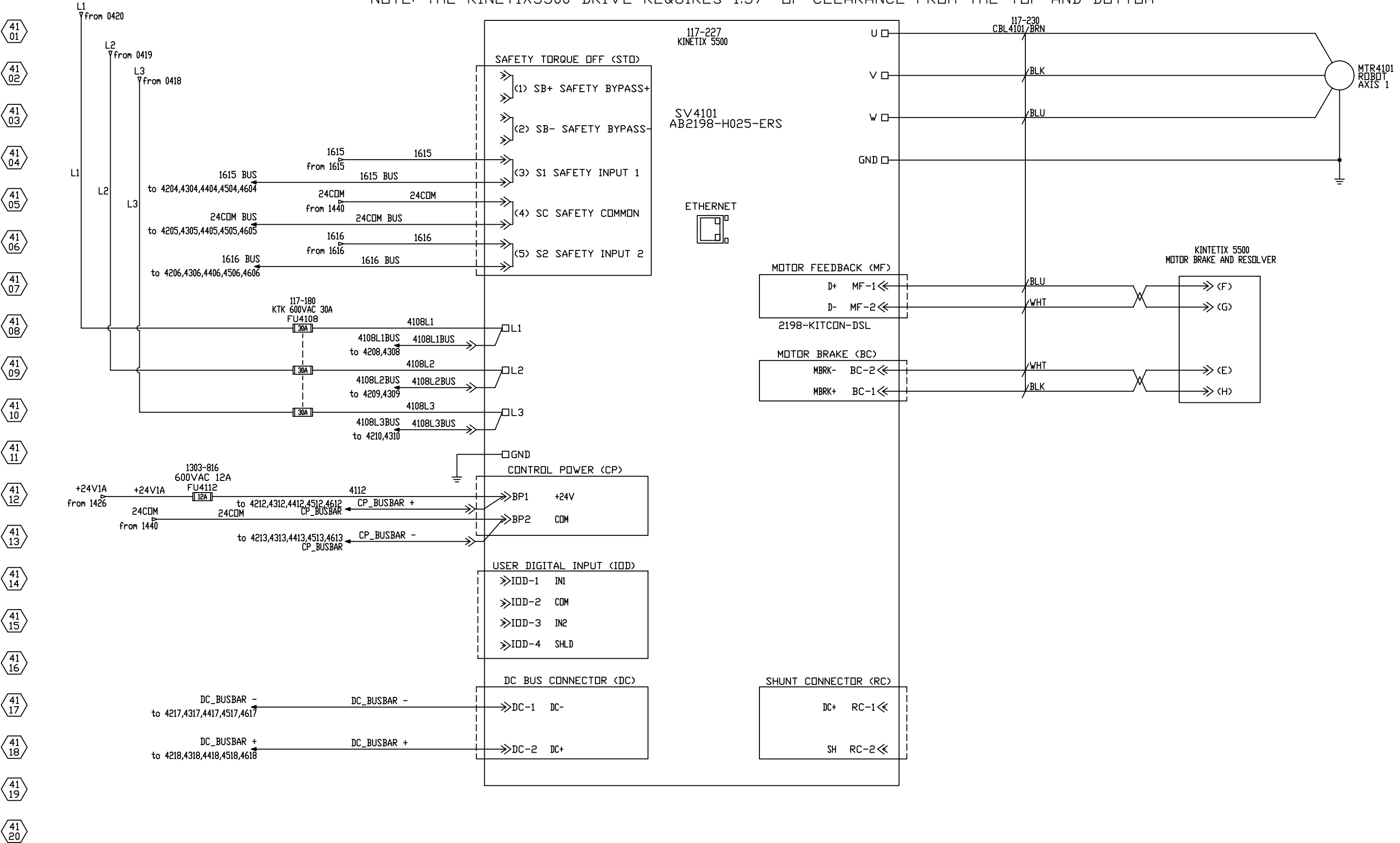


39 01
39 02
39 03
39 04
39 05
39 06
39 07
39 08
39 09
39 10
39 11
39 12
39 13
39 14
39 15
39 16
39 17
39 18
39 19
39 20

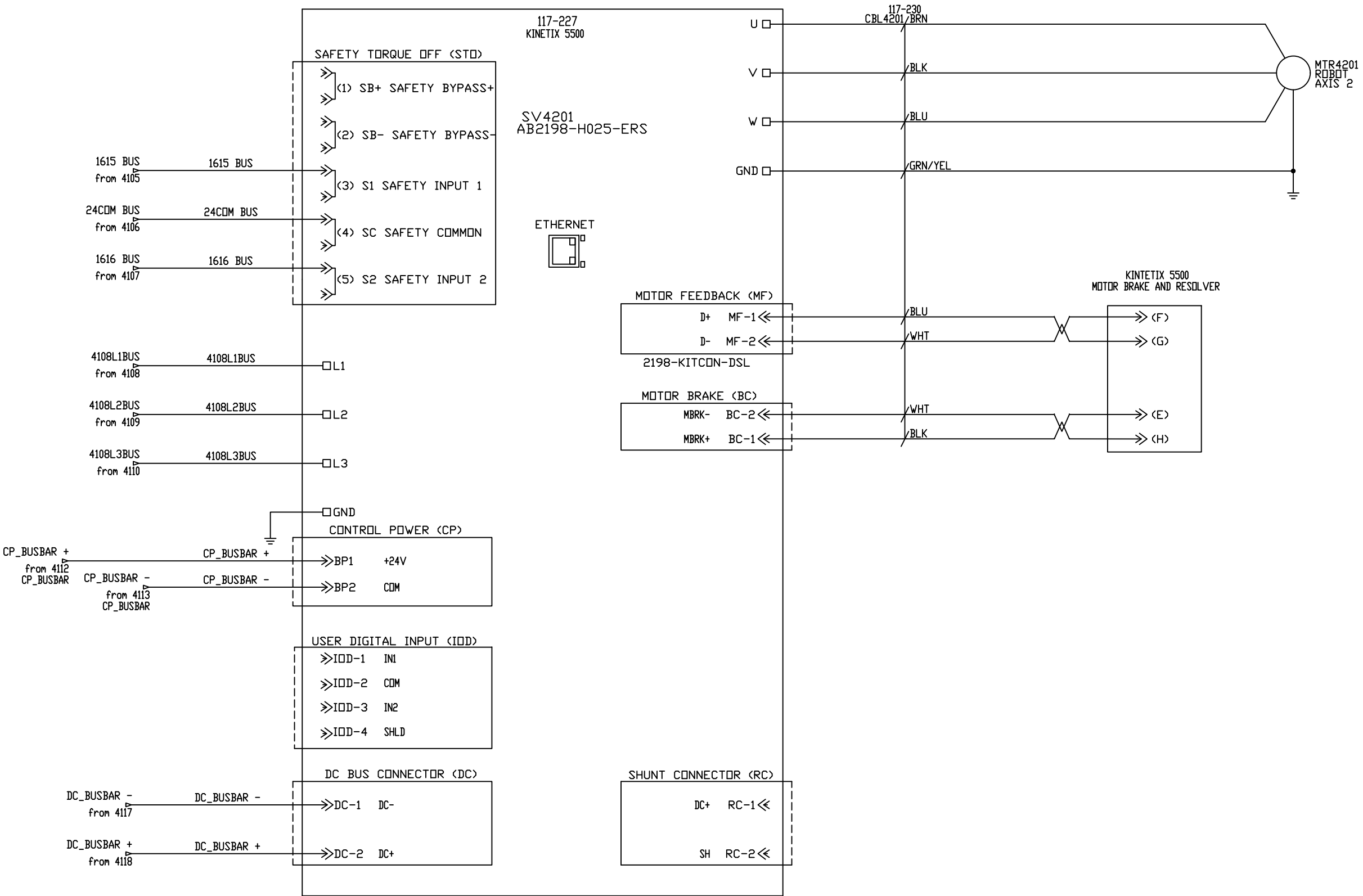
PAGES 39 & 40 WERE INTENTIONALLY LEFT BLANK

39 21
39 22
39 23
39 24
39 25
39 26
39 27
39 28
39 29
39 30
39 31
39 32
39 33
39 34
39 35
39 36
39 37
39 38
39 39
39 40

NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

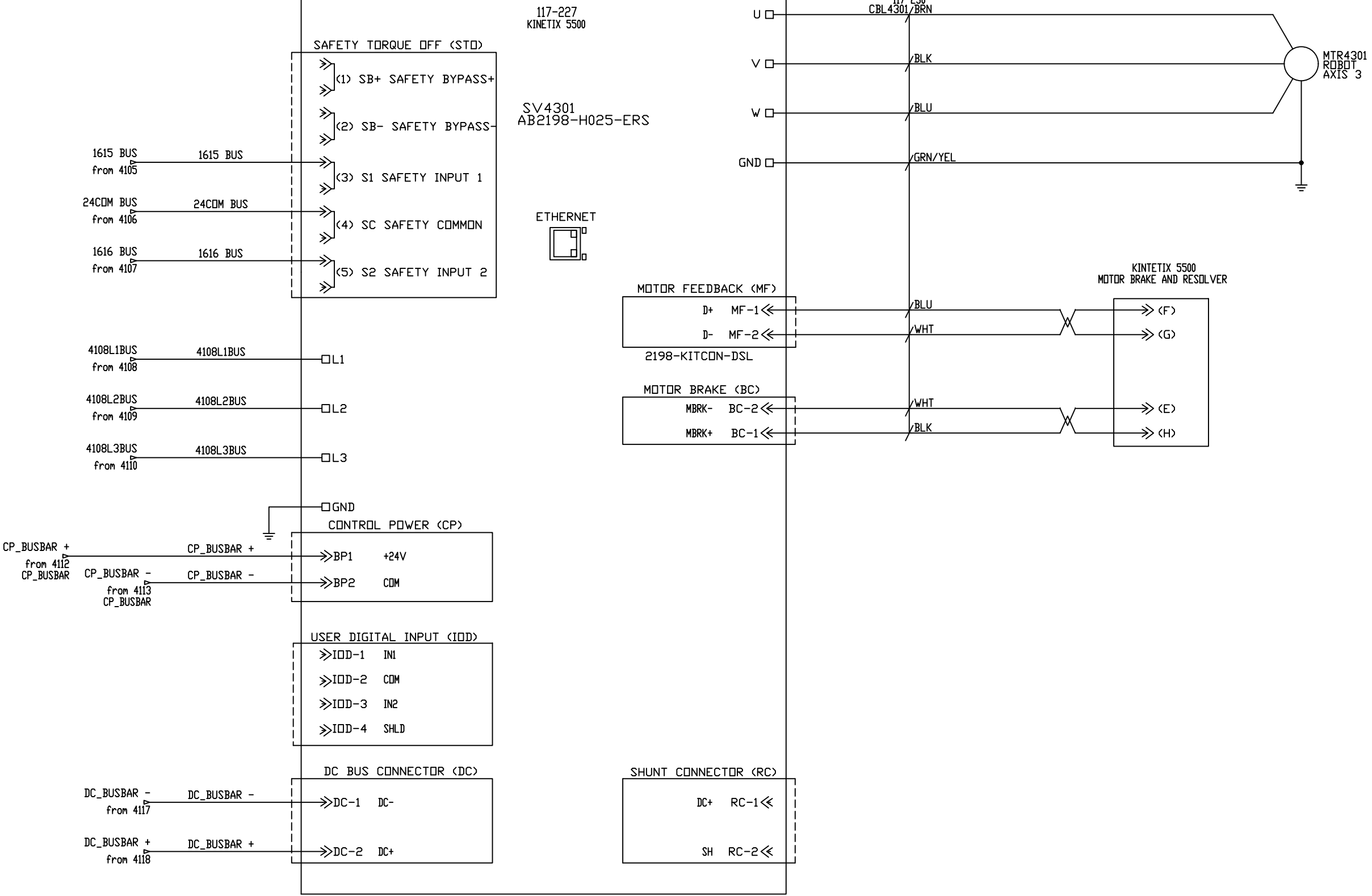


NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

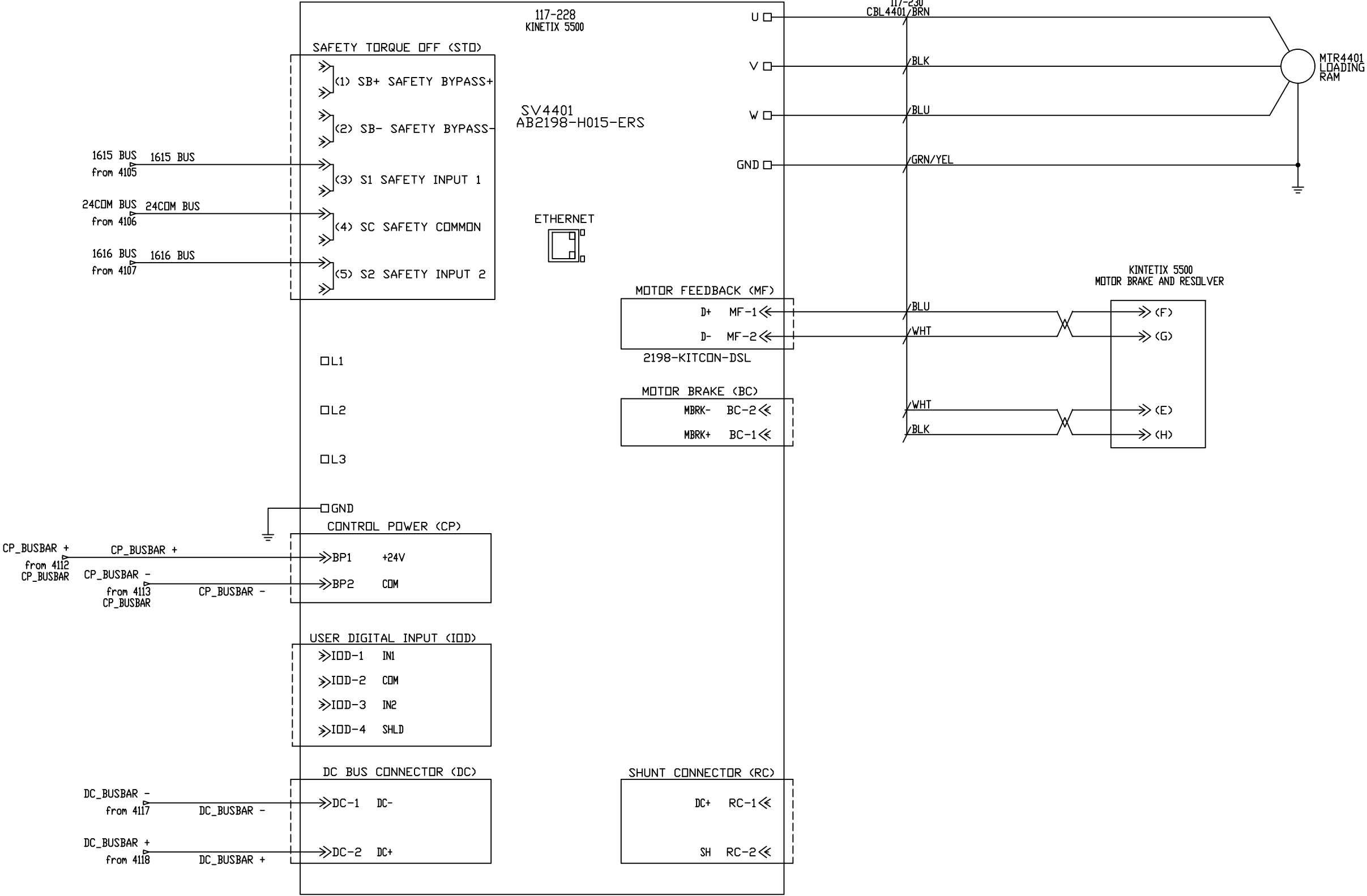


NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

43 01
43 02
43 03
43 04
43 05
43 06
43 07
43 08
43 09
43 10
43 11
43 12
43 13
43 14
43 15
43 16
43 17
43 18
43 19
43 20

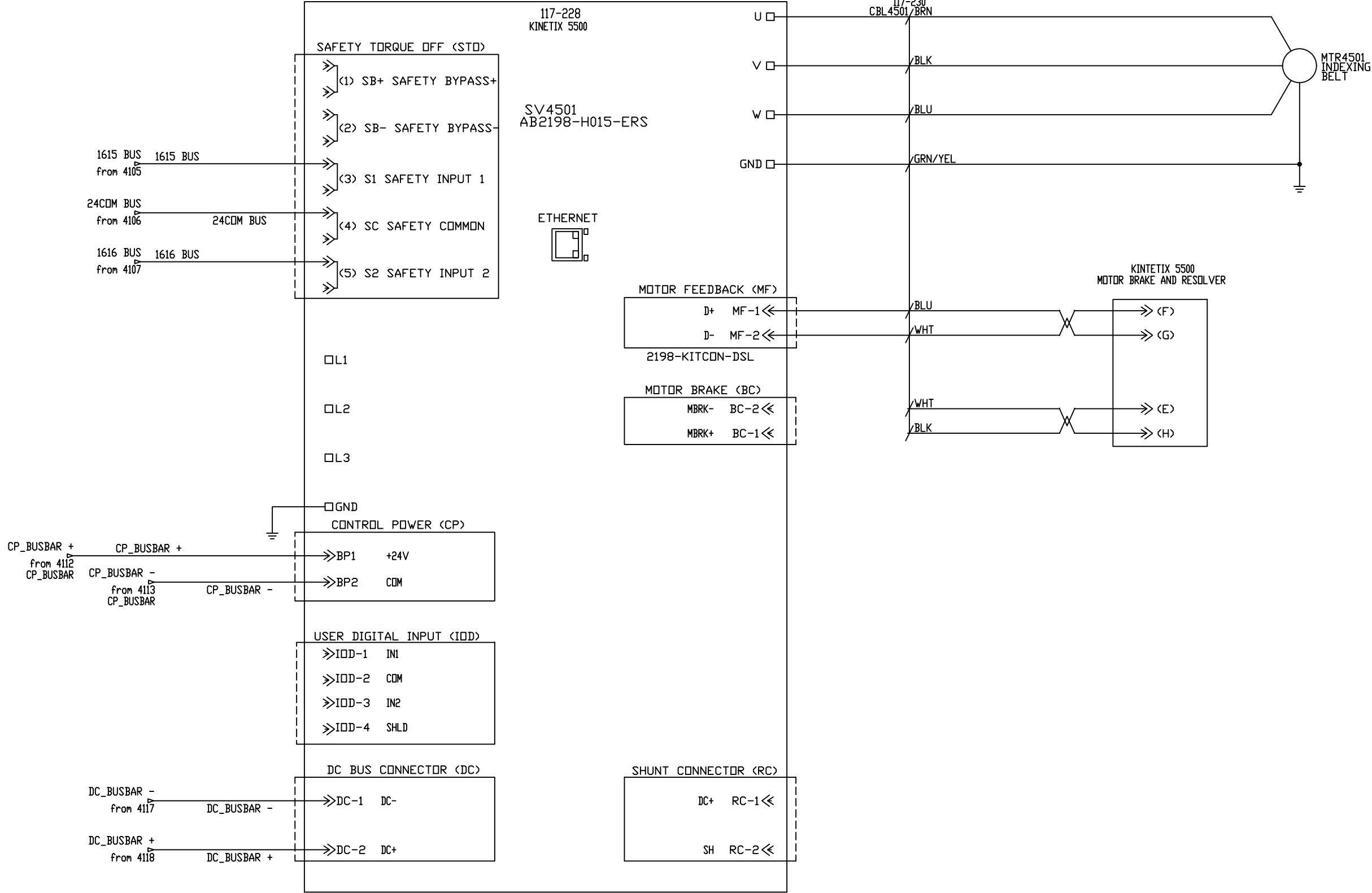


NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM



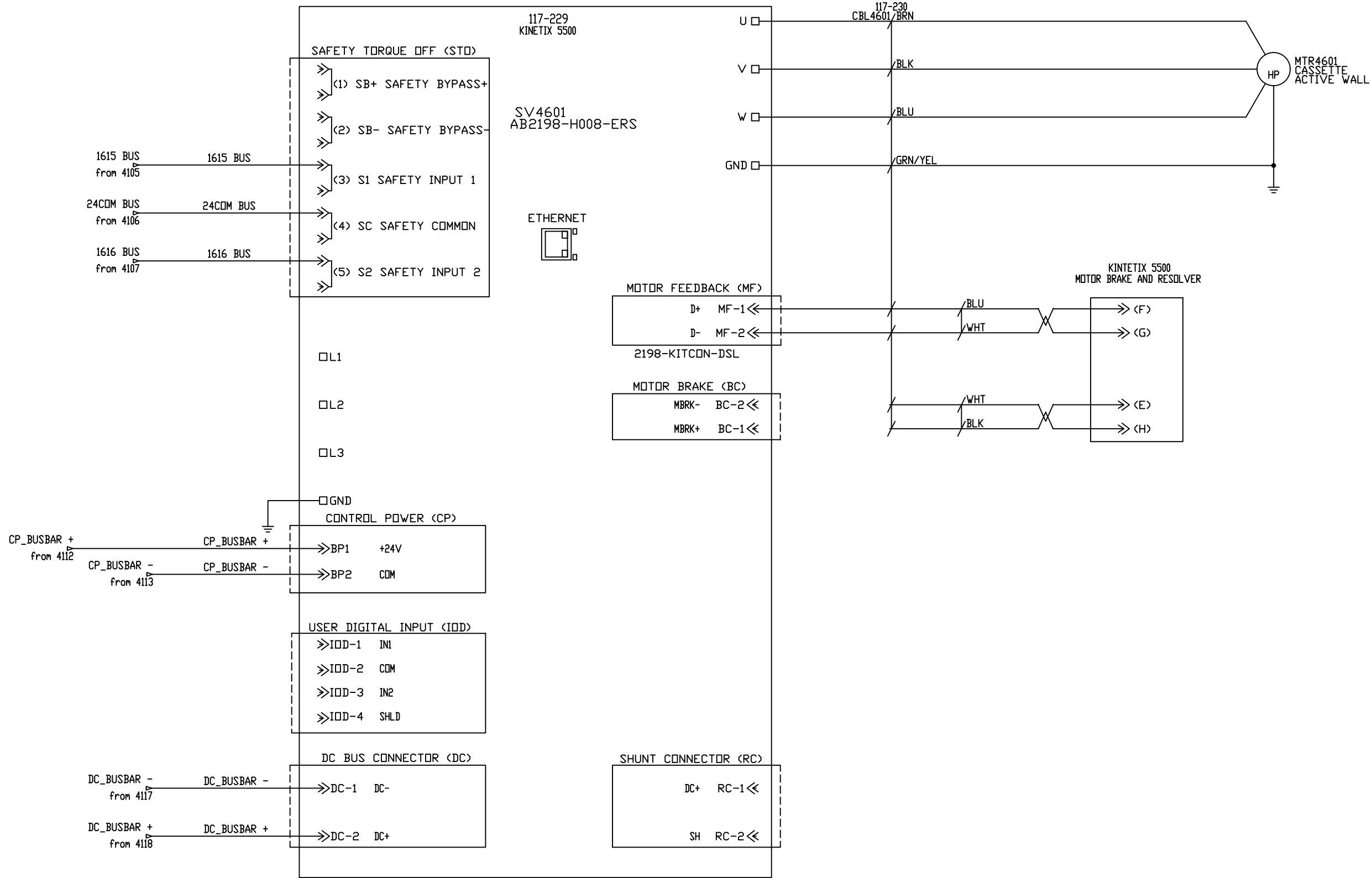
NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

45 01
45 02
45 03
45 04
45 05
45 06
45 07
45 08
45 09
45 10
45 11
45 12
45 13
45 14
45 15
45 16
45 17
45 18
45 19
45 20

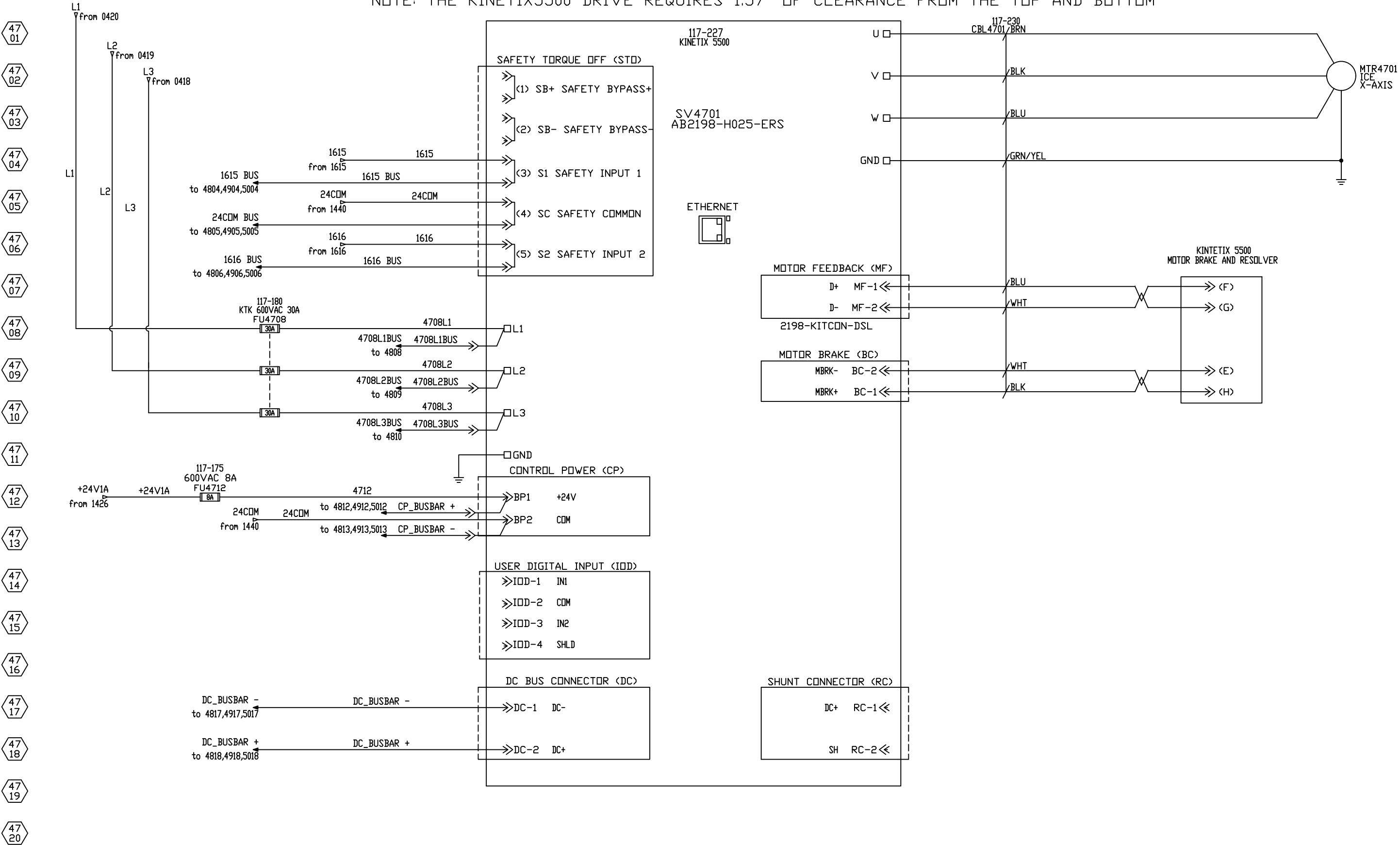


NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

46 01
46 02
46 03
46 04
46 05
46 06
46 07
46 08
46 09
46 10
46 11
46 12
46 13
46 14
46 15
46 16
46 17
46 18
46 19
46 20

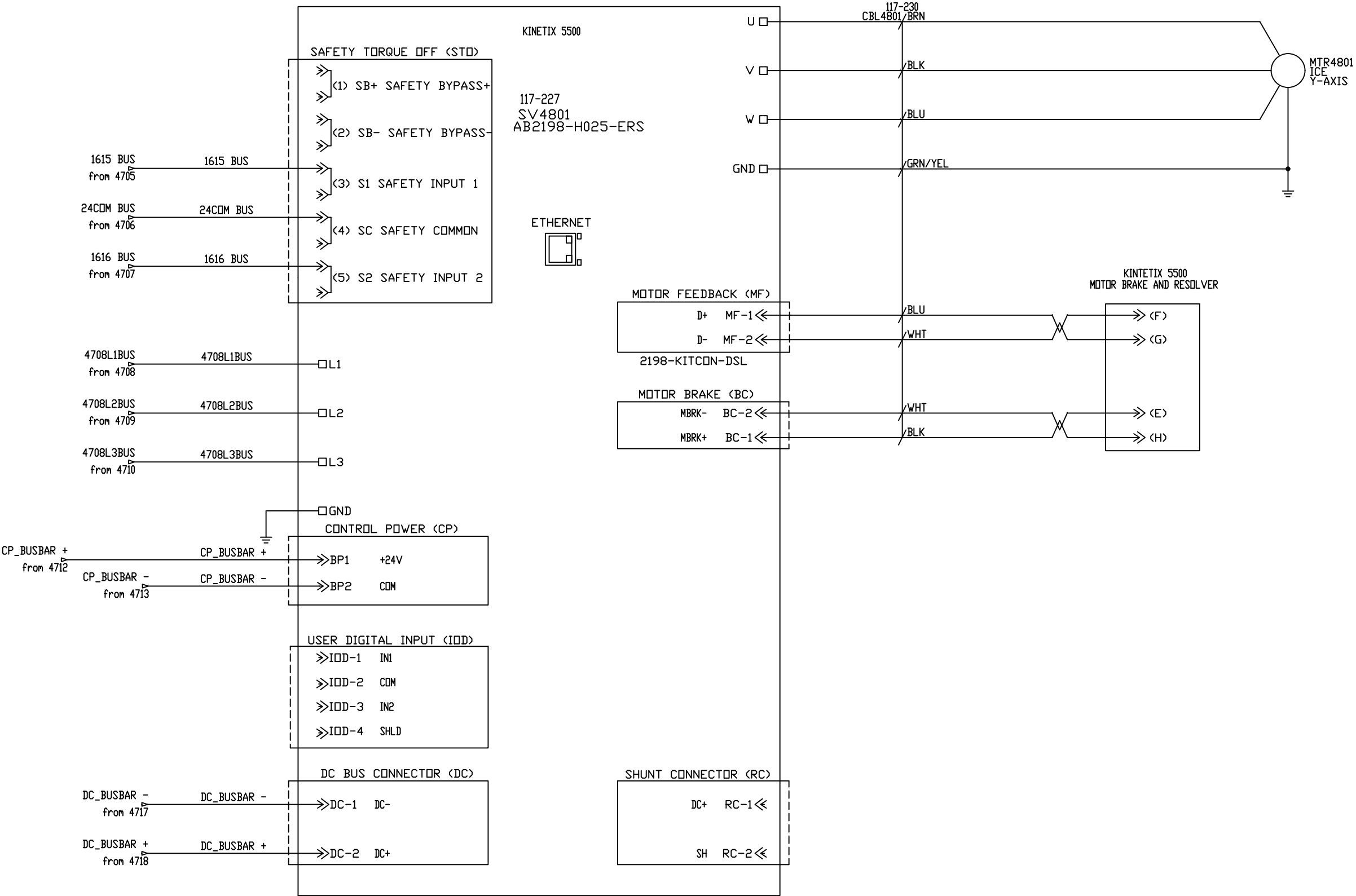


NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

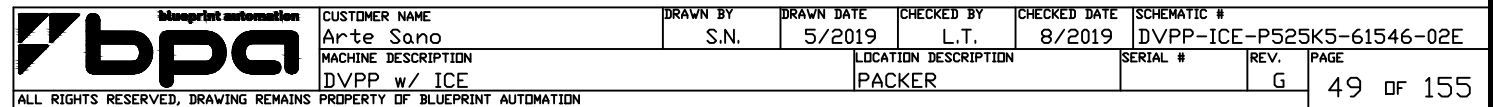


NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

48 01
48 02
48 03
48 04
48 05
48 06
48 07
48 08
48 09
48 10
48 11
48 12
48 13
48 14
48 15
48 16
48 17
48 18
48 19
48 20



- 49
01
- 49
02
- 49
03
- 49
04
- 49
05
- 49
06
- 49
07
- 49
08
- 49
09
- 49
10
- 49
11
- 49
12
- 49
13
- 49
14
- 49
15
- 49
16
- 49
17
- 49
18
- 49
19
- 49
20



NOTE: THE KINETIX5500 DRIVE REQUIRES 1.57" OF CLEARANCE FROM THE TOP AND BOTTOM

50
01

50
02

50
03

50
04

50
05

50
06

50
07

50
08

50
09

50
10

50
11

50
12

50
13

50
14

50
15

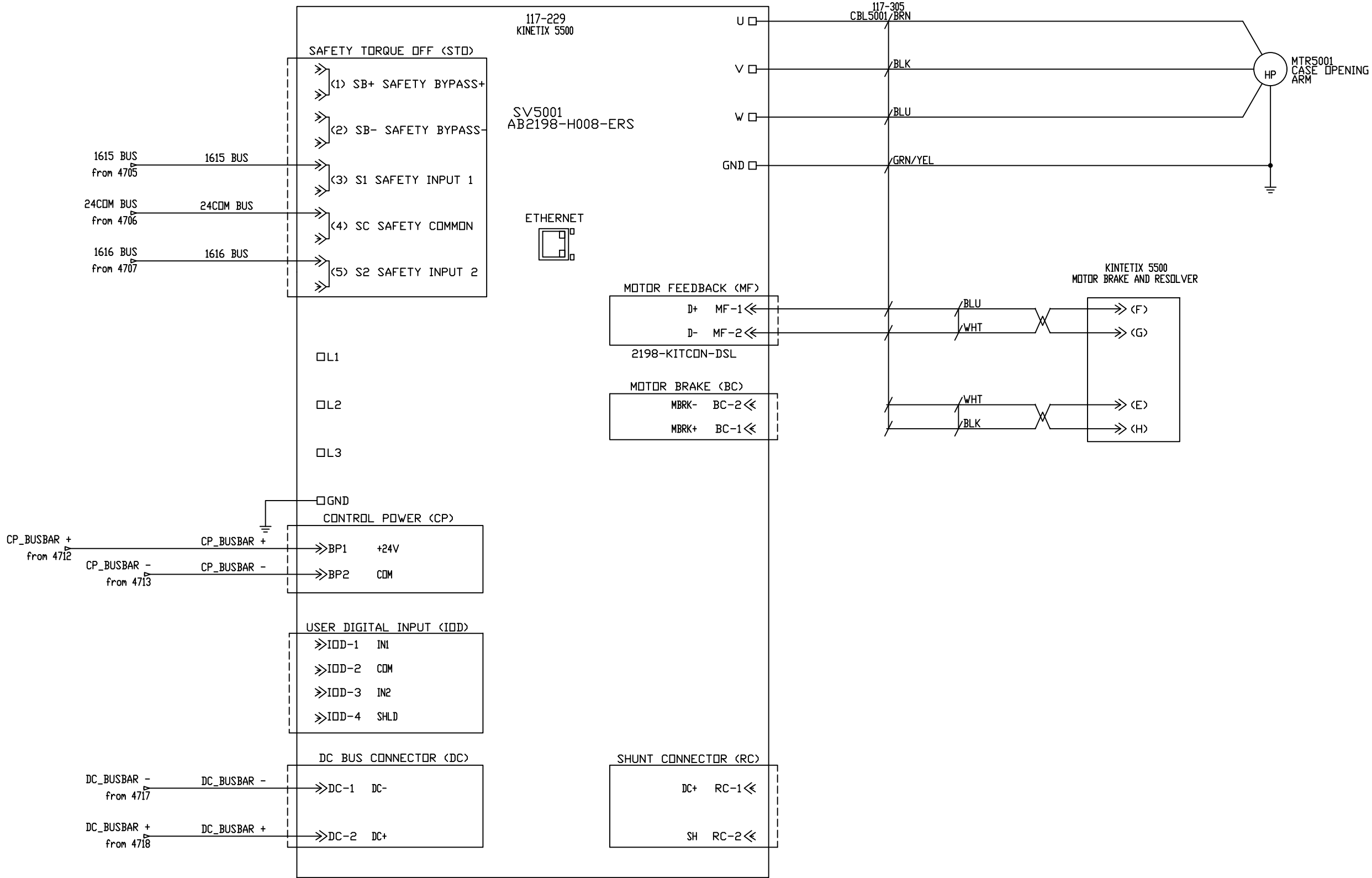
50
16

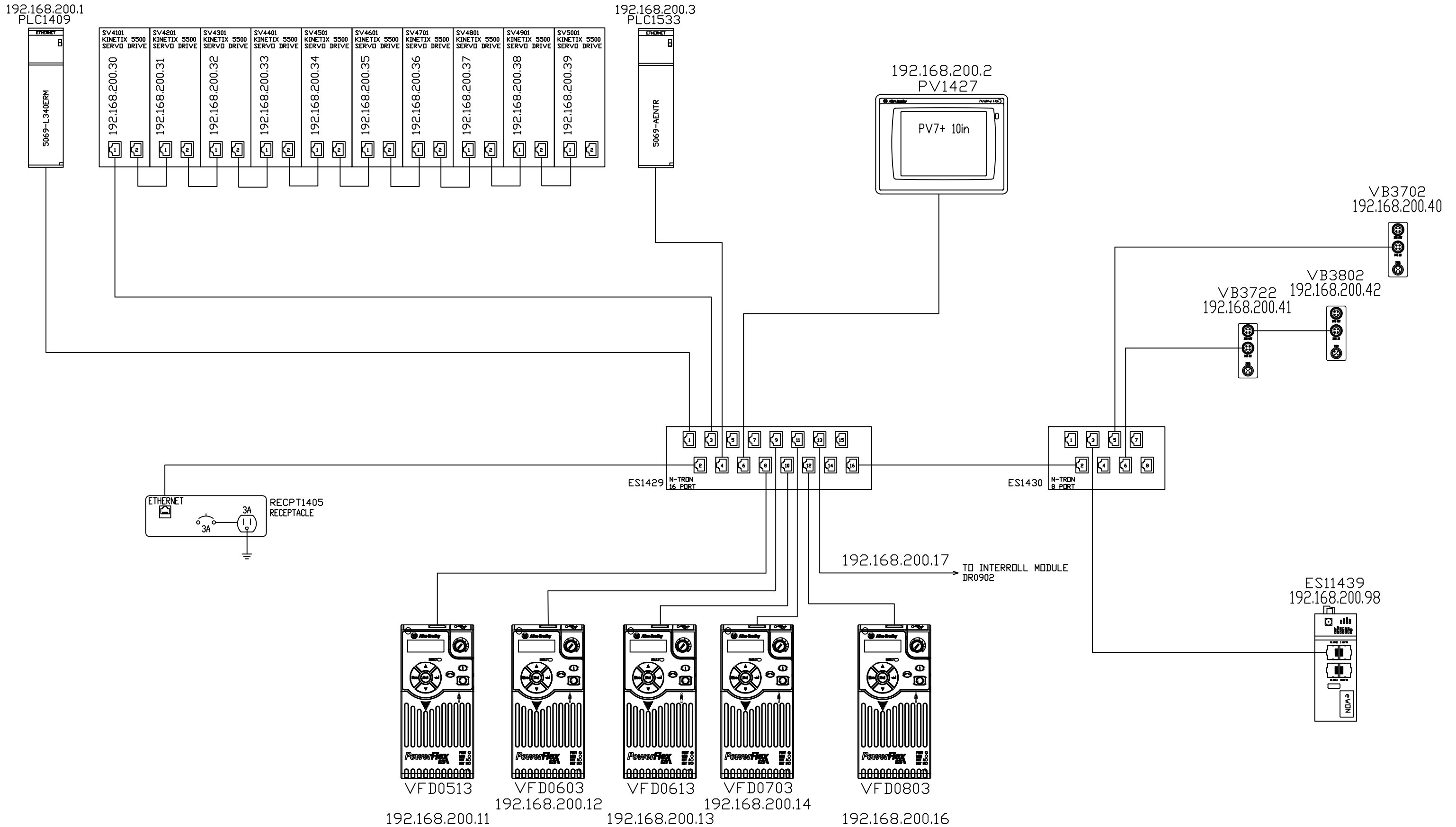
50
17

50
18

50
19

50
20





- 52 01
- 52 02
- 52 03
- 52 04
- 52 05
- 52 06
- 52 07
- 52 08
- 52 09
- 52 10
- 52 11
- 52 12
- 52 13
- 52 14
- 52 15
- 52 16
- 52 17
- 52 18
- 52 19
- 52 20

PAGES 52 - 54 WERE INTENTIONALLY LEFT BLANK

Electrical Power Component Torque Specifications

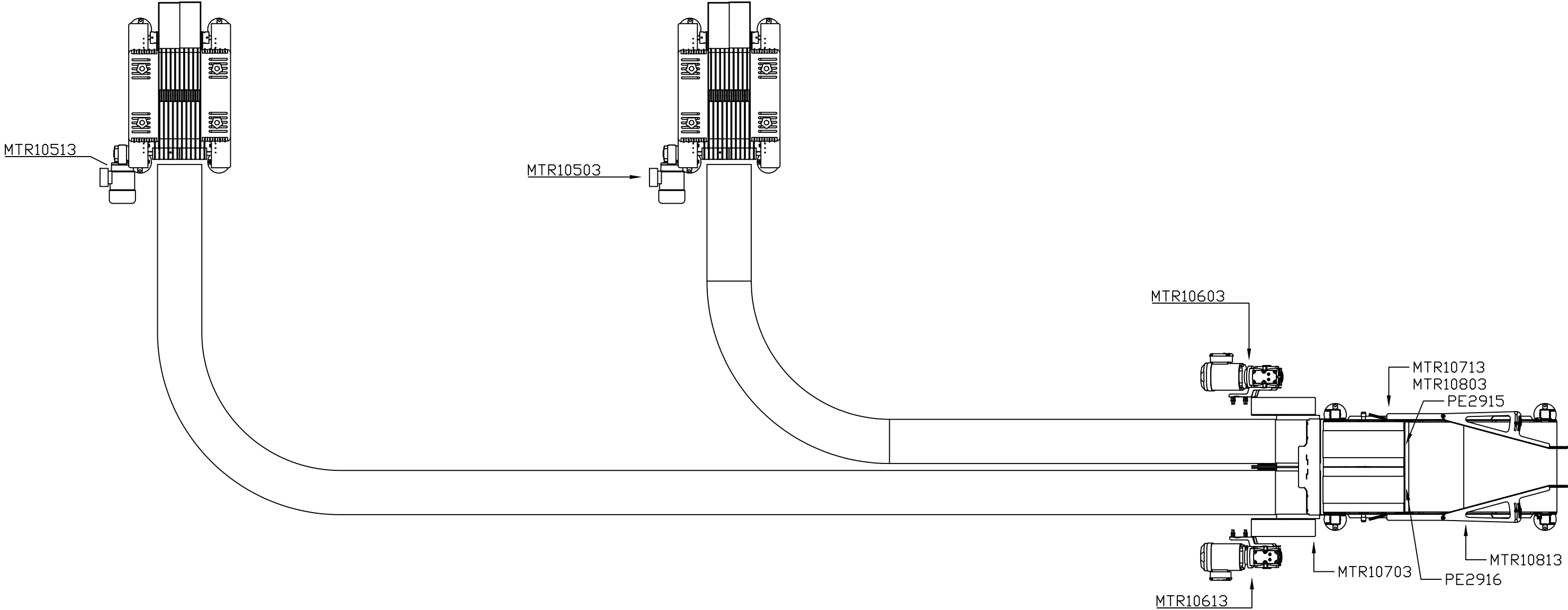
Mfg	Mfg Number	Description	Torque (Nm)	Torque (lb-in)
AB	194RF-NC030R# 194RF-NJ030R# 194RF-NN030R# 194R-NA100P3 194R-NA200P3	Disconnect, Rotary, 30A	1.4Nm	12 lb-in
AB	194RF-NJ060R# 194RF-NN060R# 194RF-NA300P3	Disconnect, Rotary, 60A	4.0Nm	35lb-in
AB	100S-C30DJ22C	Contactor, Safety	1.5-3.5Nm	13-31lb-in
AB	1492-PD3C141	Distribution Block, 3 Phase	none provided	
AB	140MCWTE	Terminal, 3Phase Feeder	3.0-3.5Nm	27-31lb-in
AB	140MC2EB63	Motor Starter, 4.0-6.3A	2.0-2.5Nm	18-22lb-in
AB	140MC2EB25	Motor Starter, 1.6-2.5A	2.0-2.5Nm	18-22lb-in
AB	140MC2EB16	Motor Starter, 1.0-1.6A	2.0-2.5Nm	18-22lb-in
AB	140MC2EB10	Motor Starter, 0.63-1.0A	2.0-2.5Nm	18-22lb-in
AB	100C12UZJ10	Contactor, Motor	1.0-2.5Nm	8.9-22lb-in
Mennekes	20MS1A-M2	Disconnect, Rotary, Non-Fused, 25A	none provided	
AB	22AD2P3N104	Drive, Inverter, PowerFlex 4M	1.7-2.2Nm	14-20lb-in
AB	22BD2P3N104	Drive, Inverter, PowerFlex 525	1.7-2.2Nm	14-20lb-in
ACME	T-2A-53340-IS	Transformer, 3 Phase, 480/240	none provided	
AB	2094-BC01-M01-S	Drive, Servo, Kinetix 6000	2.26Nm	20lb-in
Micron	B750BTZ13JK	Transformer, 1 Phase,	none provided	
Weldmuller	991549	Receptacle	none provided	
Hoffman	T-FP41	Cooling Fan	none provided	
AB	1769-PA4	Power Supply, Compact Logix, 4A	1.27Nm	11.24lb-in
AB	1606-XL240E	Power Supply, 240VAC to 24VDC	0.8Nm	7.0lb-in
Mersen	USCC1I	Fuse Holder, 1 Pole, Class CC	1.7Nm	14lb-in
Mersen	USCC2I	Fuse Holder, 2 Pole, Class CC	1.7Nm	14lb-in
Mersen	USCC3I	Fuse Holder, 3 Pole, Class CC	1.7Nm	14lb-in



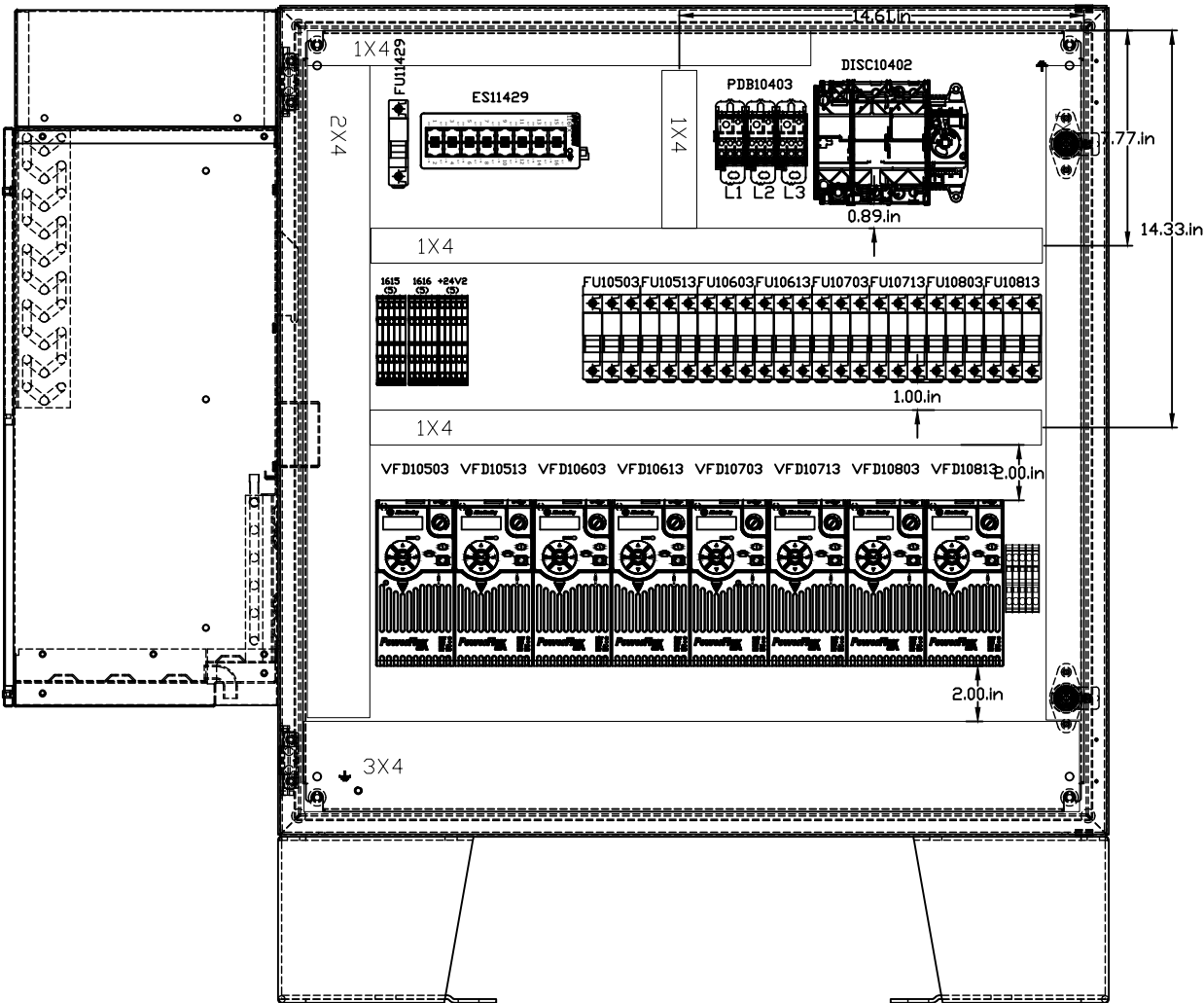
- 5601
- 5602
- 5603
- 5604
- 5605
- 5606
- 5607
- 5608
- 5609
- 5610
- 5611
- 5612
- 5613
- 5614
- 5615
- 5616
- 5617
- 5618
- 5619
- 5620

PAGES 56 - 100 WERE INTENTIONALLY LEFT BLANK

61546 Arte Sano Merge Device Layout



61546 Arte Sano Merge Device Layout



For all 24VDC fuse holders, Use: Mersen USCC1I-DC24
For all other fuse holders, Use: Mersen USCC1I, USCC2I, or USCC3I

Minimum Wire Size
Branch Circuits: 12 AWG.
External Control Circuits: 16 AWG.
Internal Control Circuits: 18 AWG.

Internal 24VDC wiring:
a. Blue (+24VDC)
b. Blue/White Stripe (common)

Power Wiring (480) - Black colored insulation
a. Black L1 (phase A)
b. Black L2 (phase B)
c. Black L3 (phase C)
d. Gray (Neutral/Grounded)

Power Wiring (120 VAC, 1 Phase)
a. Red (Phase A)
b. White (Neutral/Grounded)

103
01

103
02

103
03

103
04

103
05

103
06

103
07

103
08

103
09

103
10

103
11

103
12

103
13

103
14

103
15

103
16

103
17

103
18

103
19

103
20

103
21

103
22

103
23

103
24

103
25

103
26

103
27

103
28

103
29

103
30

103
31

103
32

103
33

103
34

103
35

103
36

103
37

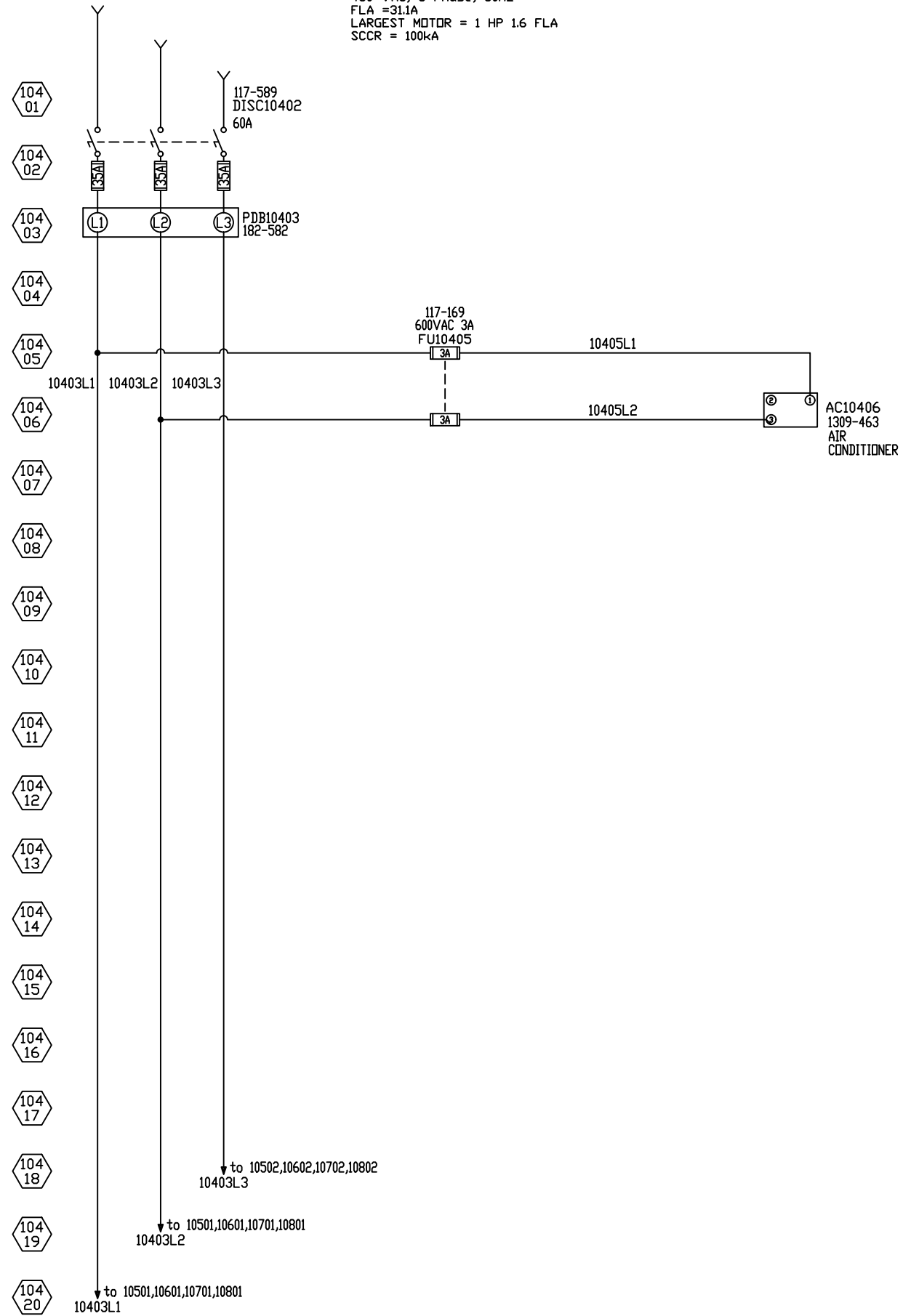
103
38

103
39

103
40

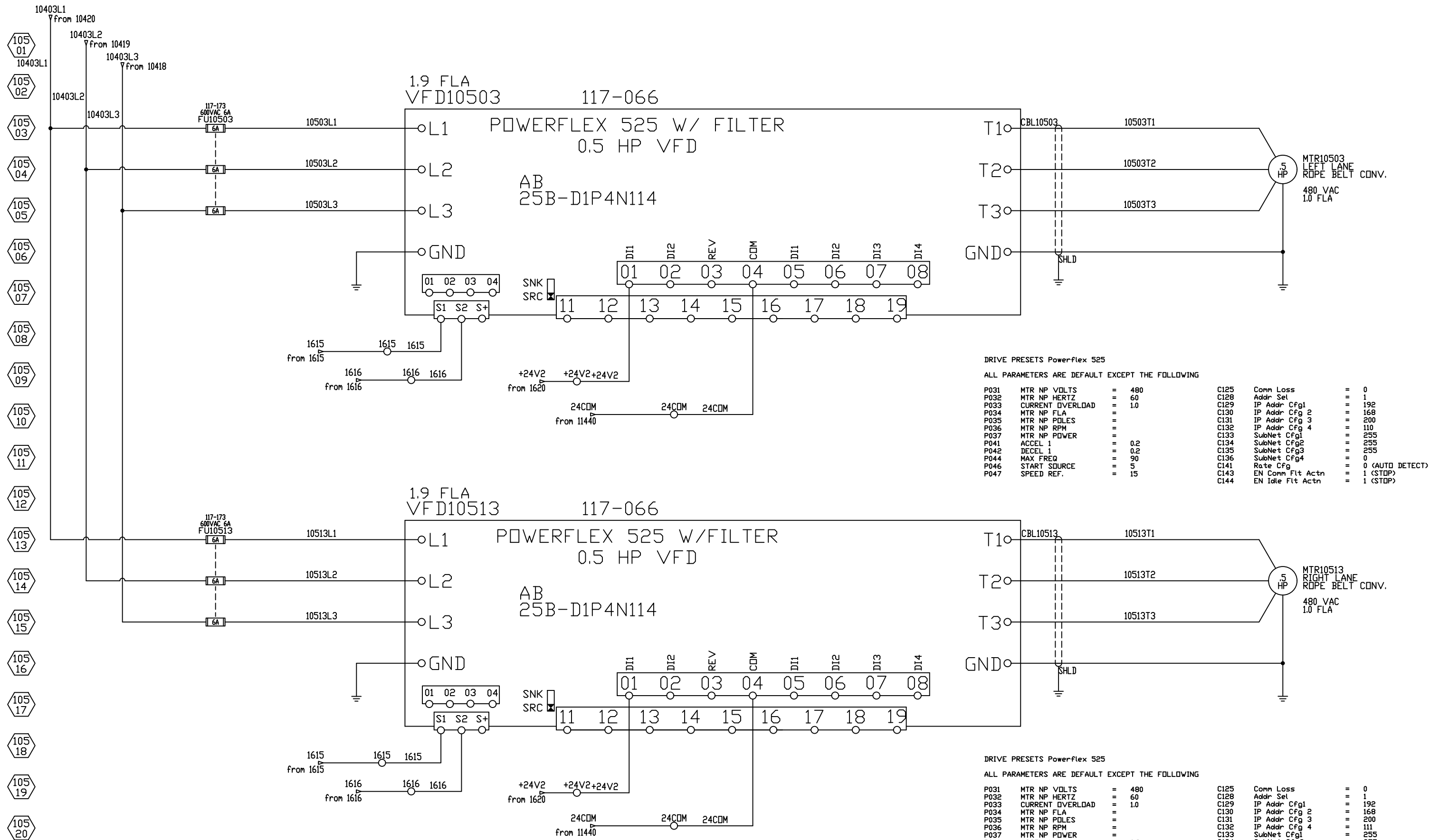
PAGE 103 WAS INTENTIONALLY LEFT BLANK

480 VAC, 3 Phase, 60hz
FLA =31.1A
LARGEST MOTOR = 1 HP 1.6 FLA
SCCR = 100kA



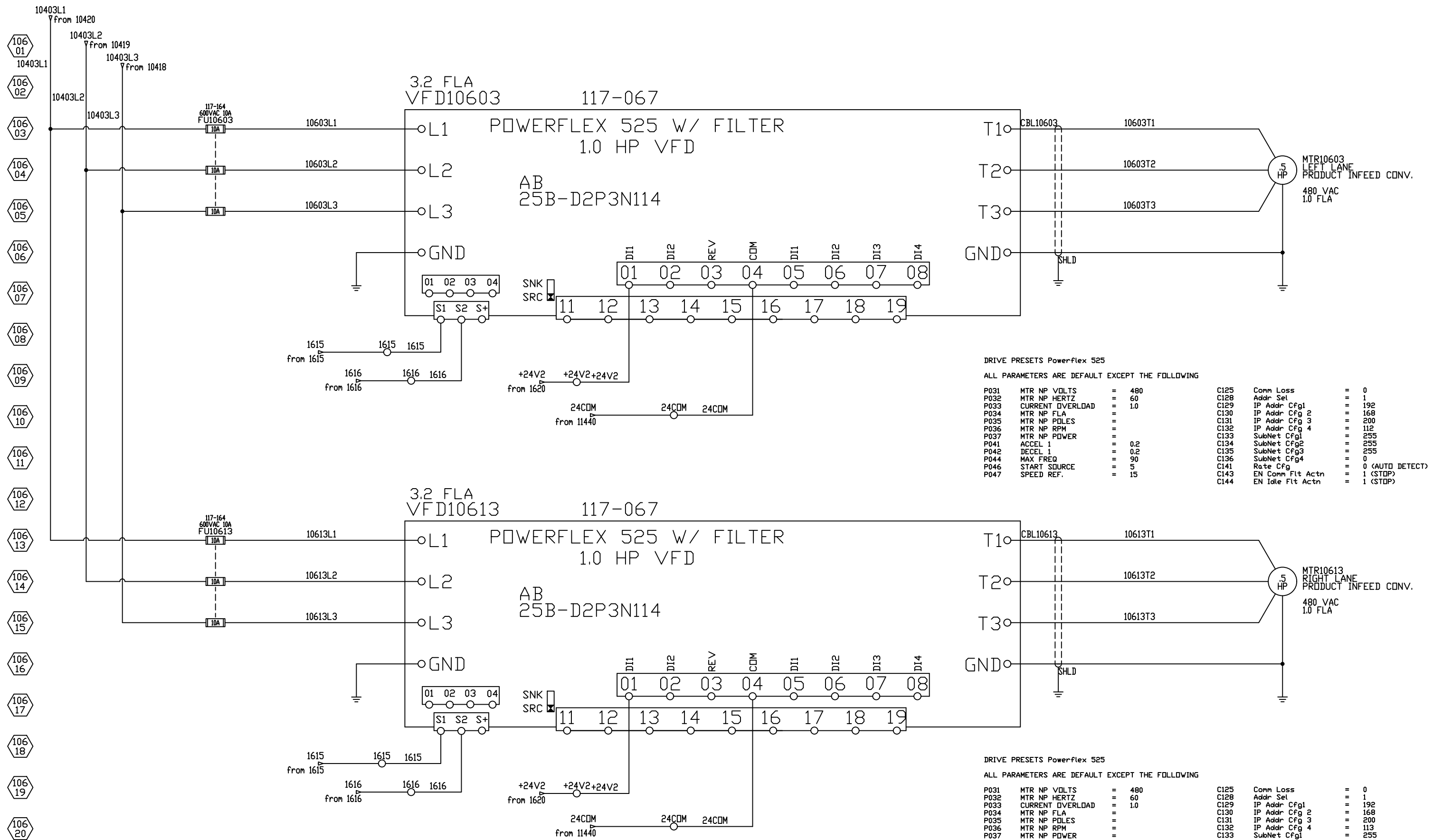
CUSTOMER NAME	Arte Sano	DRAWN BY	S.N.	DRAWN DATE	5/2019	CHECKED BY	L.T.	CHECKED DATE	8/2019	SCHEMATIC #	DVPP-ICE-P525K5-61546-02E
MACHINE DESCRIPTION	DVPP w/ ICE	LOCATION DESCRIPTION	Merge Cabinet	SERIAL #		REV.	G	PAGE	104	OF	155

ALL RIGHTS RESERVED, DRAWING REMAINS PROPERTY OF BLUEPRINT AUTOMATION



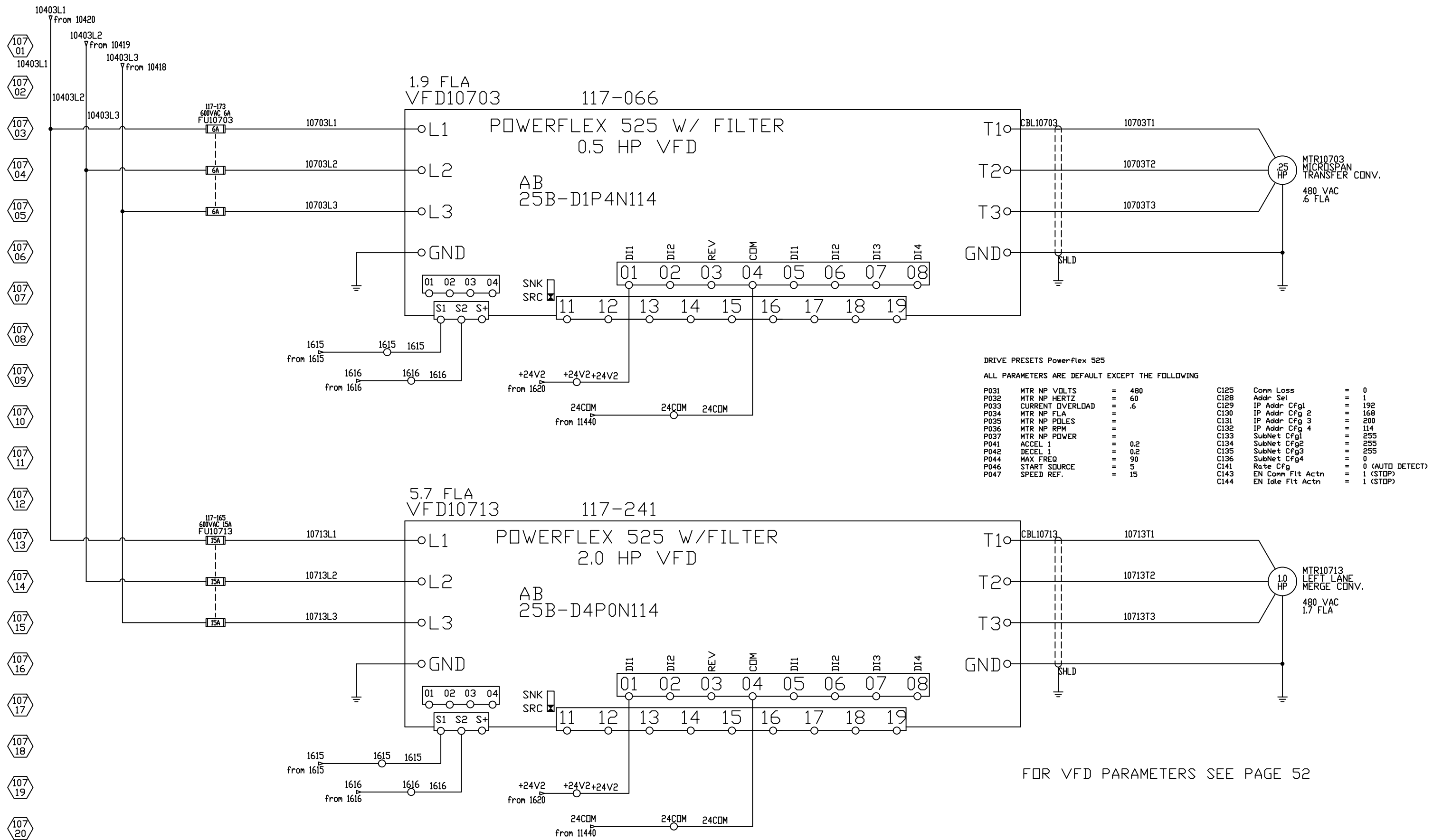
Safe Torque Setup for Replacement Drive

1. Loosen the screw of terminals Safety 1, Safety 2 and Safety +24V (S1, S2, S+) on the control I/O terminal block.
2. Remove the protective jumper.
3. Safe-Torque-Off function is now enabled.



Safe Torque Setup for Replacement Drive

1. Loosen the screw of terminals Safety 1, Safety 2 and Safety +24V (S1, S2, S+) on the control I/O terminal block.
2. Remove the protective jumper.
3. Safe-Torque-Off function is now enabled.



DRIVE PRESETS Powerflex 525

ALL PARAMETERS ARE DEFAULT EXCEPT THE FOLLOWING

P031	MTR NP VOLTS	=	480	C125	Comm Loss	=	0
P032	MTR NP HERTZ	=	60	C128	Addr Sel	=	1
P033	CURRENT OVERLOAD	=	.6	C129	IP Addr Cfg1	=	192
P034	MTR NP FLA	=		C130	IP Addr Cfg 2	=	168
P035	MTR NP POLES	=		C131	IP Addr Cfg 3	=	200
P036	MTR NP RPM	=		C132	IP Addr Cfg 4	=	114
P037	MTR NP POWER	=		C133	SubNet Cfg1	=	255
P041	ACCEL 1	=	0.2	C134	SubNet Cfg2	=	255
P042	DECEL 1	=	0.2	C135	SubNet Cfg3	=	255
P044	MAX FREQ	=	90	C136	SubNet Cfg4	=	0
P046	START SOURCE	=	5	C141	Rate Cfg	=	0 (AUTO DETECT)
P047	SPEED REF.	=	15	C143	EN Comm Fit Actn	=	1 (STOP)
				C144	EN Idle Fit Actn	=	1 (STOP)

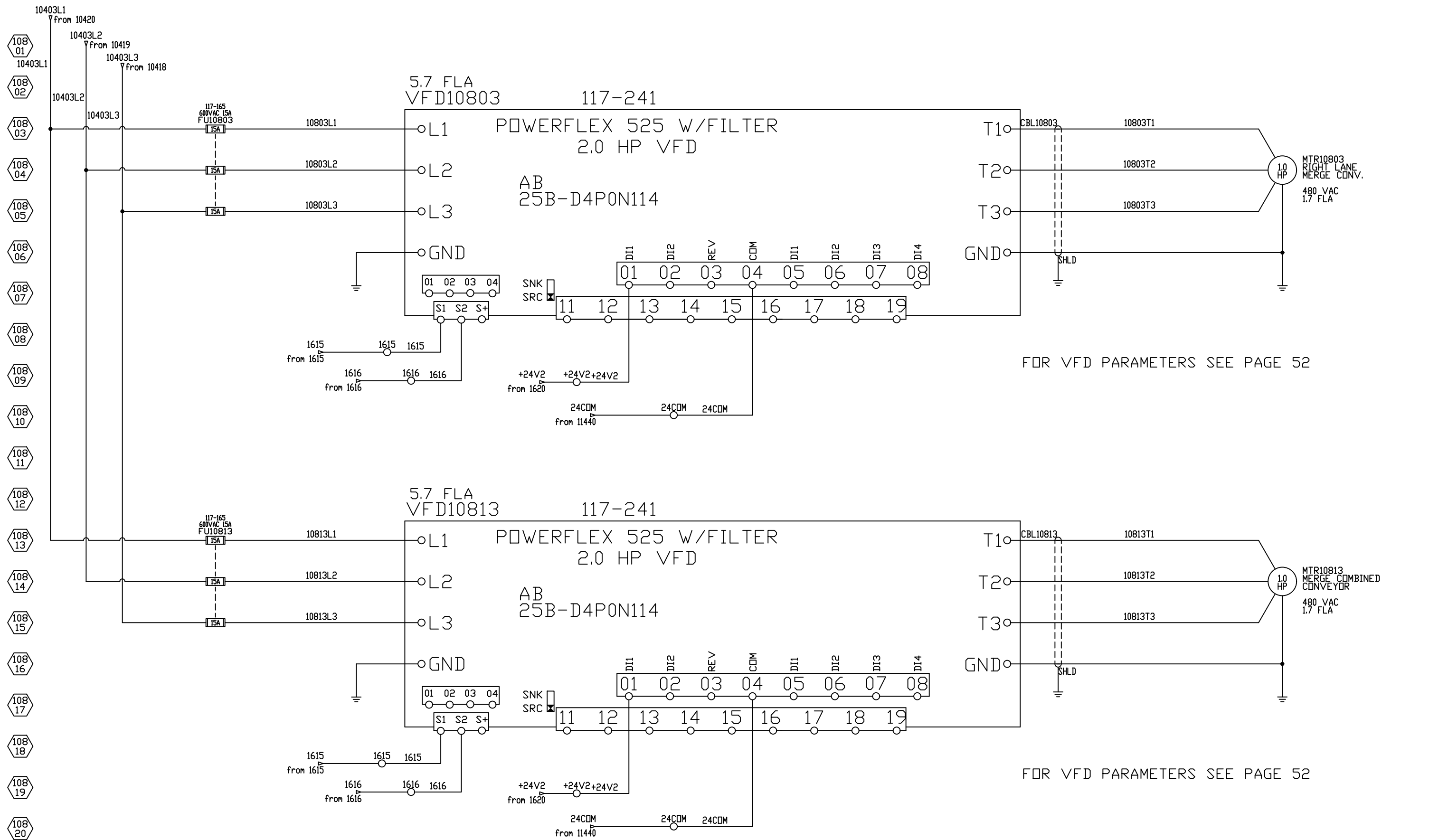
Safe Torque Setup for Replacement Drive

1. Loosen the screw of terminals Safety 1, Safety 2 and Safety +24V (S1, S2, S+) on the control I/O terminal block.
2. Remove the protective jumper.
3. Safe-Torque-Off function is now enabled.



CUSTOMER NAME	Arte Sano	DRAWN BY	S.N.	DRAWN DATE	5/2019	CHECKED BY	L.T.	CHECKED DATE	8/2019	SCHEMATIC #	DVPP-ICE-P525K5-61546-02E
MACHINE DESCRIPTION	DVPP w/ ICE	LOCATION DESCRIPTION	Merge Cabinet	SERIAL #		REV.	G	PAGE	107	OF	155

ALL RIGHTS RESERVED, DRAWING REMAINS PROPERTY OF BLUEPRINT AUTOMATION



Safe Torque Setup for Replacement Drive

1. Loosen the screw of terminals Safety 1, Safety 2 and Safety +24V (S1, S2, S+) on the control I/O terminal block.
2. Remove the protective jumper.
3. Safe-Torque-Off function is now enabled.



ALL RIGHTS RESERVED, DRAWING REMAINS PROPERTY OF BLUEPRINT AUTOMATION

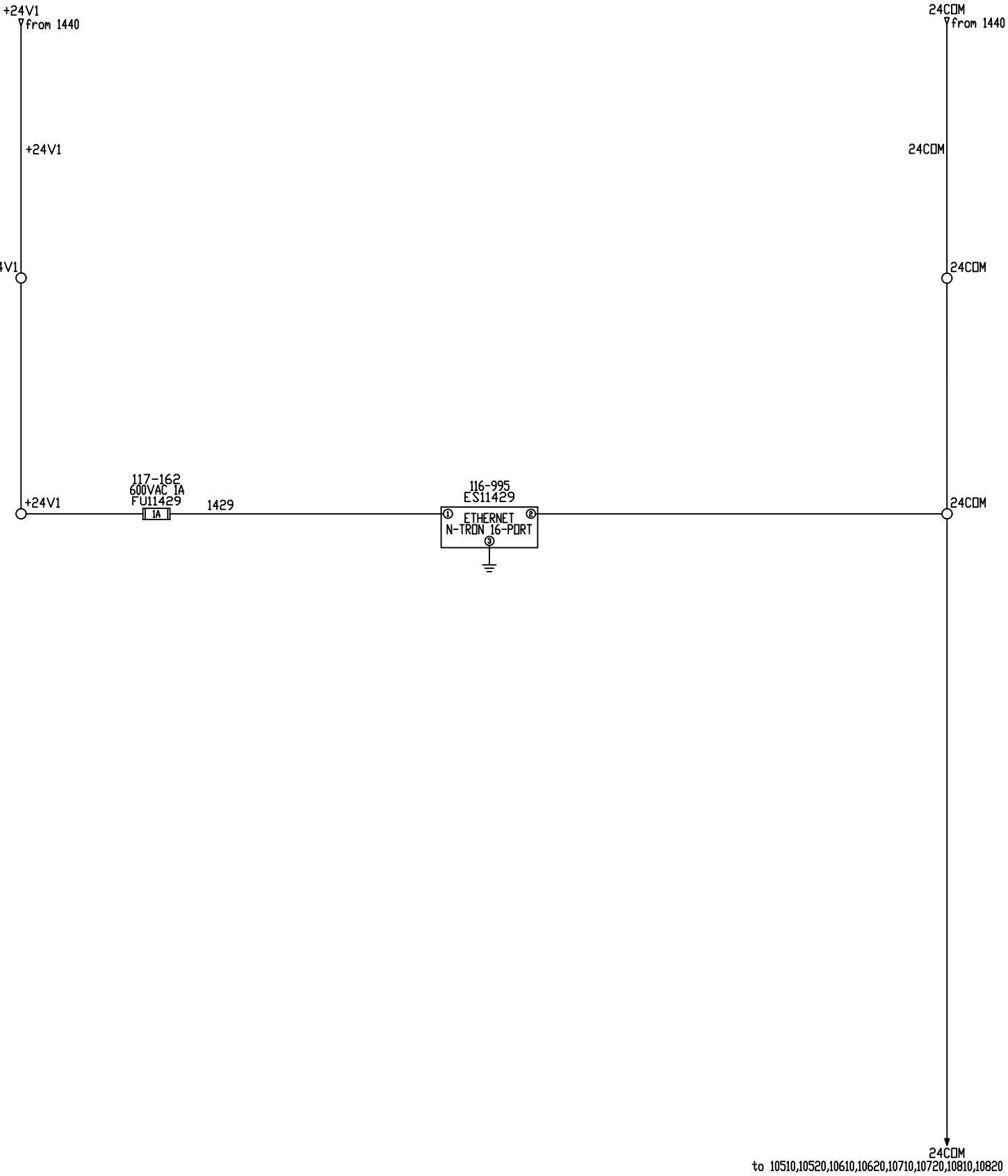
CUSTOMER NAME		DRAWN BY	DRAWN DATE	CHECKED BY	CHECKED DATE	SCHEMATIC #	
Arte Sano		S.N.	5/2019	L.T.	8/2019	DVPP-ICE-P525K5-61546-02E	
MACHINE DESCRIPTION		LOCATION DESCRIPTION		SERIAL #		REV.	PAGE
DVPP w/ ICE		Merge Cabinet				G	108 of 155

- 109 01
- 109 02
- 109 03
- 109 04
- 109 05
- 109 06
- 109 07
- 109 08
- 109 09
- 109 10
- 109 11
- 109 12
- 109 13
- 109 14
- 109 15
- 109 16
- 109 17
- 109 18
- 109 19
- 109 20

PAGES 109 - 113 WERE INTENTIONALLY LEFT BLANK

- 114 01
- 114 02
- 114 03
- 114 04
- 114 05
- 114 06
- 114 07
- 114 08
- 114 09
- 114 10
- 114 11
- 114 12
- 114 13
- 114 14
- 114 15
- 114 16
- 114 17
- 114 18
- 114 19
- 114 20

- 114 21
- 114 22
- 114 23
- 114 24
- 114 25
- 114 26
- 114 27
- 114 28
- 114 29
- 114 30
- 114 31
- 114 32
- 114 33
- 114 34
- 114 35
- 114 36
- 114 37
- 114 38
- 114 39
- 114 40



115
01

115
02

115
03

115
04

115
05

115
06

115
07

115
08

115
09

115
10

115
11

115
12

115
13

115
14

115
15

115
16

115
17

115
18

115
19

115
20

PAGES 115 - 149 WERE INTENTIONALLY LEFT BLANK

115
21

115
22

115
23

115
24

115
25

115
26

115
27

115
28

115
29

115
30

115
31

115
32

115
33

115
34

115
35

115
36

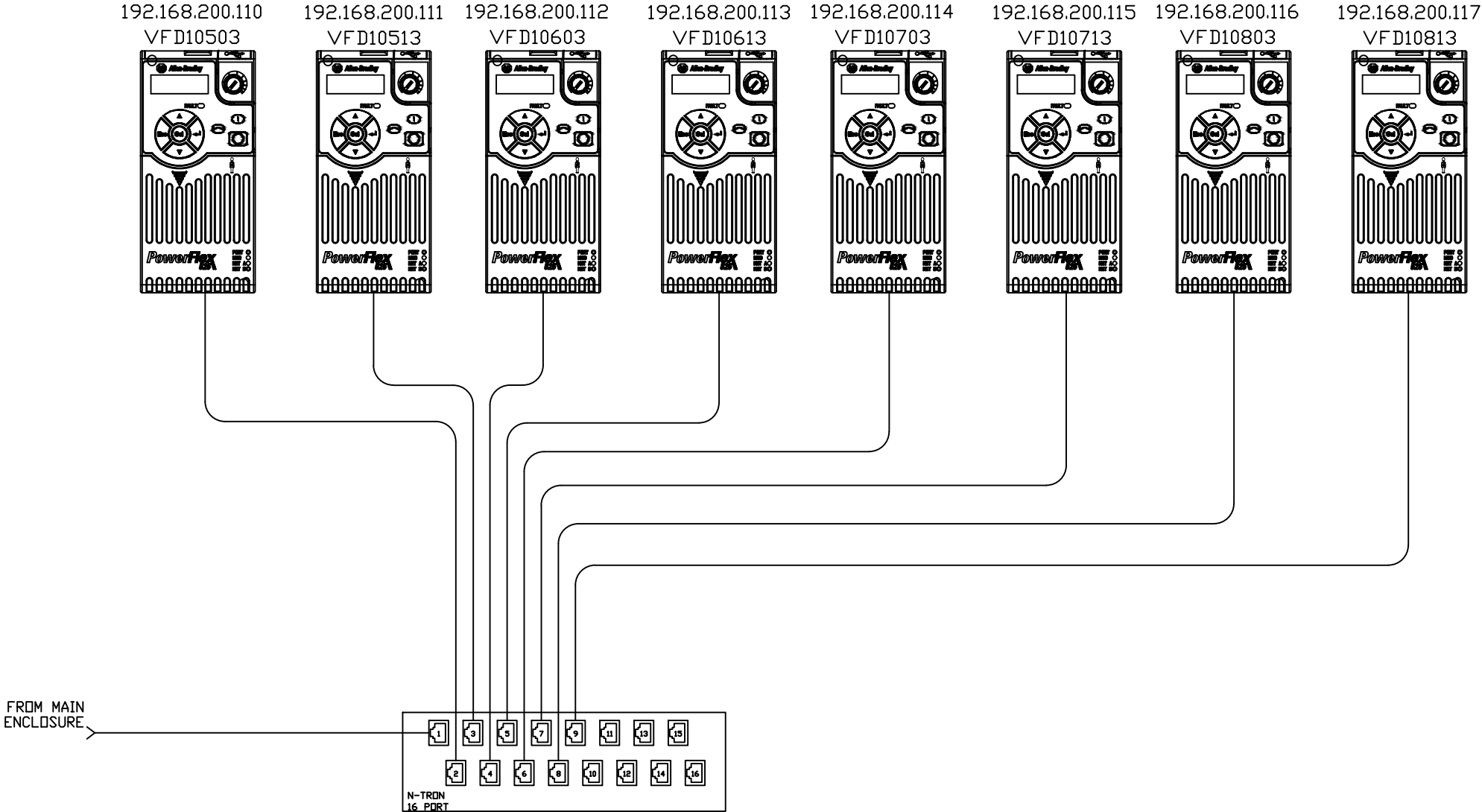
115
37

115
38

115
39

115
40

- 150 01
- 150 02
- 150 03
- 150 04
- 150 05
- 150 06
- 150 07
- 150 08
- 150 09
- 150 10
- 150 11
- 150 12
- 150 13
- 150 14
- 150 15
- 150 16
- 150 17
- 150 18
- 150 19
- 150 20



- 151
01
- 151
02
- 151
03
- 151
04
- 151
05
- 151
06
- 151
07
- 151
08
- 151
09
- 151
10
- 151
11
- 151
12
- 151
13
- 151
14
- 151
15
- 151
16
- 151
17
- 151
18
- 151
19
- 151
20

PAGES 151 - 154 WERE INTENTIONALLY LEFT BLANK

Electrical Power Component Torque Specifications

Mfg	Mfg Number	Description	Torque (Nm)	Torque (lb-in)
AB	194RF-NC030R* 194RF-NJ030R* 194RF-NN030R* 194R-NA100P3 194R-NA200P3	Disconnect, Rotary, 30A	1.4Nm	12 lb-in
AB	194RF-NJ060R* 194RF-NN060R* 194RF-NA300P3	Disconnect, Rotary, 60A	4.0Nm	35lb-in
AB	100SC37ZJ14C	Contactor, Safety	1.5-3.5Nm	13-31lb-in
AB	1492-PD3C141	Distribution Block, 3 Phase	none provided	
AB	140MCWTE	Terminal, 3Phase Feeder	3.0-3.5Nm	27-31lb-in
AB	140MC2EB63	Motor Starter, 4.0-6.3A	2.0-2.5Nm	18-22lb-in
AB	140MC2EB25	Motor Starter, 1.6-2.5A	2.0-2.5Nm	18-22lb-in
AB	140MC2EB16	Motor Starter, 1.0-1.6A	2.0-2.5Nm	18-22lb-in
AB	140MC2EB10	Motor Starter, 0.63-1.0A	2.0-2.5Nm	18-22lb-in
AB	100C12UZJ10	Contactor, Motor	1.0-2.5Nm	8.9-22lb-in
Mennekes	20MS1A-M2	Disconnect, Rotary, Non-Fused, 25A	none provided	
AB	1492-FB3C30-L	Fuse Holder, 3 Pole, Class CC	2.0-2.5Nm	18-22lb-in
AB	22AD2P3N104	Drive, Inverter, PowerFlex4	1.7-2.2Nm	14-20lb-in
AB	22BD2P3N104	Drive, Inverter, PowerFlex40	1.7-2.2Nm	14-20lb-in
AB	1492-FB1C30-L	Fuse Holder, 1 Pole, Class CC	2.0-2.5Nm	18-22lb-in
ACME	T-2A-53340-1S	Transformer, 3 Phase, 480/240	none provided	
AB	1492-FB2C30-L	Fuse Holder, 2 Pole, Class CC	2.0-2.5Nm	18-22lb-in
AB	2094-BC01-M01-S	Drive, Servo, Kinetix 6000	2.26Nm	20lb-in
Micron	B750BTZ13JK	Transformer, 1 Phase,	none provided	
Weidmuller	991549	Receptacle	none provided	
Hoffman	T-FP41	Cooling Fan	none provided	
AB	1769-PA4	Power Supply, Compact Logix, 4A	1.27Nm	11.24lb-in
AB	1606-XL240E	Power Supply, 240VAC to 24VDC	0.8Nm	7.0lb-in

ALL RIGHTS RESERVED, DRAWING REMAINS PROPERTY OF BLUEPRINT AUTOMATION